Everdirect®



Better Chemistry
Better Life

Everlight Chemical

臺灣永光化學工業股份有限公司

Everlight Chemical Industrial Corporation

Everdirect

中文版

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Better Chemistry Better Life

品牌承諾

永光化學追求進步創新，以滿載的研究熱情，透過不斷的加乘與連結，
創造出許許多多的可能。永光，以化學專業給予客戶最大支持，協助客戶邁向成功，
也為美好的社會及永續的未來，做出具體貢獻。
我們相信，運用對的化學品，可以讓生活更美好！



品牌價值

永光化學相信，運用對的化學品，可以讓生活更美好！

穩健可靠
正派經營的理念、履行承諾的決心、穩定的品質與貼心的服務，奠定永光化學的品牌信賴感。

以客為尊
專注於化學本業，滿載研究熱情，永光化學追求突破與進步，讓化學對人類的貢獻，無止境地向前演進。

進步創新
關懷顧客、共創價值是永光化學的品質政策。我們以高品質產品、服務與技術支援，給予客戶支持，協助客戶邁向成功。

造福人群
提升人類生活福祉，是永光存在的意義；以不斷進步的化學專業，為更好明天與永續未來，做出具體的貢獻。

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1.介紹：

永光化學將“Everdirect dyes”定位為高品質及產品面廣之直接染料。這些染料具備優異的染色牢度及應用特性。

Everdirect 直接染料可適用於各種的纖維素纖維，如棉、縲縈、麻等。同時這些染料對縲縈染色上可獲得良好的濕牢度。

Everdirect 直接染料依特點可歸類成以下4類：

(1) Everdirect 系列 - - - - - - - - - - - - - - - - - - - 優異之日光牢度特性
(2) Everdirect Supra系列 - - - - - - - - - - - - - - - 優異之日光及濕牢度特性
(3) Everdirect Light 系列 - - - - - - - - - - - - - - - 具備像反應性染料明亮之色彩
(4) Everdirect Cupro系列 - - - - - - - - - - - - - - 經含銅離子之固色劑固色後可獲得優異之濕牢度

2.Everdirect染料系列：

1. Everdirect Supra Yellow PG
2. Everdirect Yellow RL
3. Everdirect Supra Orange 2GL
4. Everdirect Light Scarlet F2G
5. Everdirect Supra Red BWS
6. Everdirect Light Rose FR
7. Everdirect Supra Rubine BL
8. Everdirect Brown GTL
9. Everdirect Turq. Blue FBL
10. Everdirect Turq. Blue GRL
11. Everdirect Blue FFRL
12. Everdirect Supra Blue BGL
13. Everdirect Supra Blue 4BL H/C
14. Everdirect Supra Blue BRL
15. Everdirect Supra Blue BRN
16. Everdirect Supra Blue GDL
17. Everdirect Supra Blue BRR
18. Everdirect Supra Grey CGL
19. Everdirect Black HRN
20. Everditrct Fast Black B160
21. Everdirect Fast Black VSF 600
22. Everdirect Black ANBN H/C
23. Everdirect Fast Black B300

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3.Everdirect染料一般使用方法：

Everdirect染料可適用於纖維素纖維及其混紡品以浸染、壓染及印花方式施染。

3-1. 染色素材前處理：

染色素材前處理的好壞將會直接影響染色品質。因為染色素材上的油劑及漿料必須去除完全才可以避免不均染。前處理後殘餘之化學助劑液必須完全去除以防止在染色過程造成染色物的損傷。因此，前處理的過程中的退漿、精練、漂白、絲光等工程都必須儘可能小心。

3-2. 染色方法：

於染液中加入中性鹽(芒硝或粗鹽)，再將被染色物於30-50℃下投入染浴中。然後將染液緩慢昇溫至沸點及持溫30-60分鐘。染色後將染色物以溢流洗淨避免染料滲色。最後以另一道清水浴以固色劑固色。

(1) 染料溶解步驟：

1. 準備較大的容器溶解染料
2. 以小量的水將染料化成糊狀
3. 加入熱水並攪拌一段時間
4. 確認染料已完全溶解，並在加入染浴前以濾網過濾

(2) 鹽用量推薦表:

相對之鹽用量乃是根據染色材質、染料、染色深度及浴比來決定。

Everdirect推薦之鹽用量如下:

染色深度 (o.w.f.)	
極淺色系 (低於0.05%)	0-2 g/l
淺色系 (0.05~1.0%)	2-5 g/l
中色系 (1.0~4.0%)	5-10 g/l
深色系 (4%以上)	20 g/l

上表是依浴比 1:10 計算

※ Everdirect Fast Black B-160、Fast Black B-300及Fast Black VSF 600，在染浴中加入2~4 g/l的純鹼使PH達到10.0~10.5可達到濃染效果。

(3) 固色處理：

染色後之染色物可以下列方式施以固色處理以提高濕牢度：

固色劑	：0.5~2.0 g/l
浴比	：1:10
溫度x時間	：60℃ x 20-30 分

(4) 不均染之修色：

當發現不均染發生時可以下列方式修色：

純鹼	：3~5 g/l
非離子介面活性劑	：2~3 g/l
焦磷酸鈉	：1 g/l
溫度x時間	：100℃x 30 分

使用高浴比可獲得較佳的修色效果。

(5) 剝色：

保險粉　　：3~5 g/l

片鹼　　：2 g/l

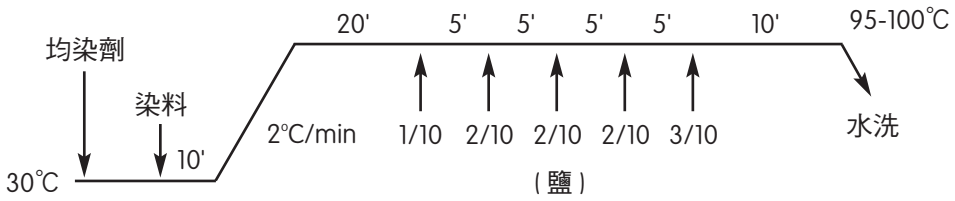
溫度x時間　　：80~90℃x 30分鐘

通常並無法完全將顏色均勻且完全剝除，而過強的剝色會造成染色物手感不良。

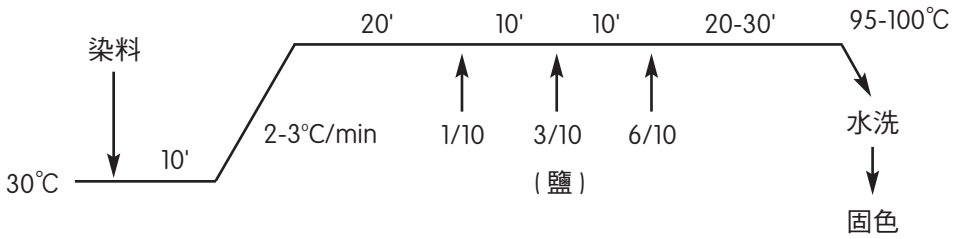
在此狀況下，最好施以適當的剝色然後改染深色。

3-3. Everdirect 染料浸染染色方法：

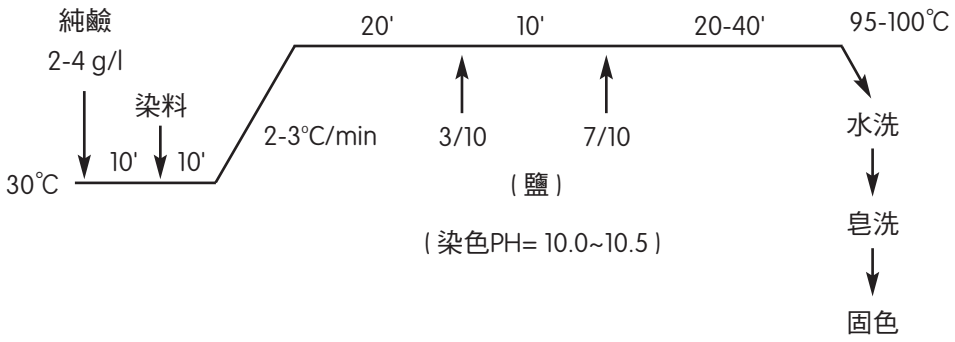
(1) 淺色系：



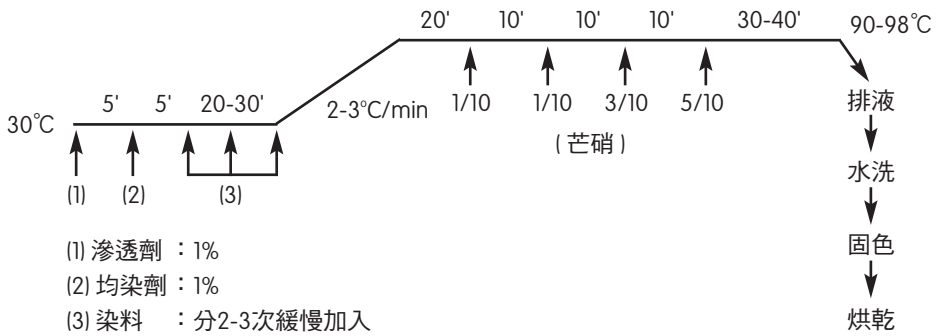
(2) 中等至深色系：



(3) Everdirect Fast Black B-160，Fast Black B-300，Fast Black VSF 600 特殊染法：



(4) Everdirect Supra FBL 及GRL染色法：



(5) 染料及鹽起始投入法：

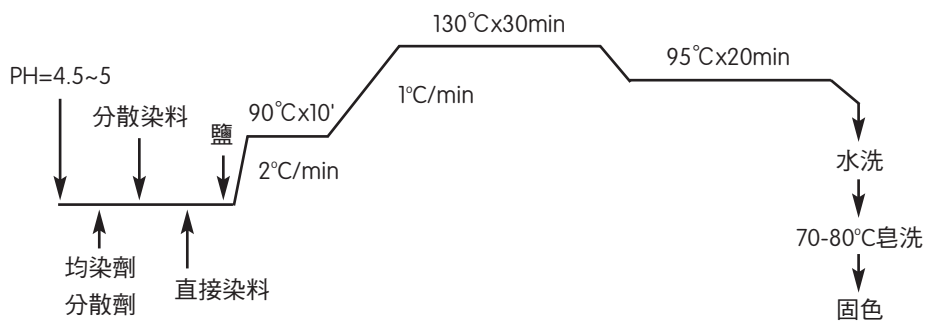


附註：

1. Blue FBL及GRL可用於T/C一浴染。
2. 染色浴比建議必須控制於1:10至1:20。
3. Blue FBL及GRL染色僅推薦使用芒硝 (不推薦使用粗鹽) 。
4. 染色後之水洗工程相當重要。水洗不足將引起不均染。另固色後必須立即烘乾。

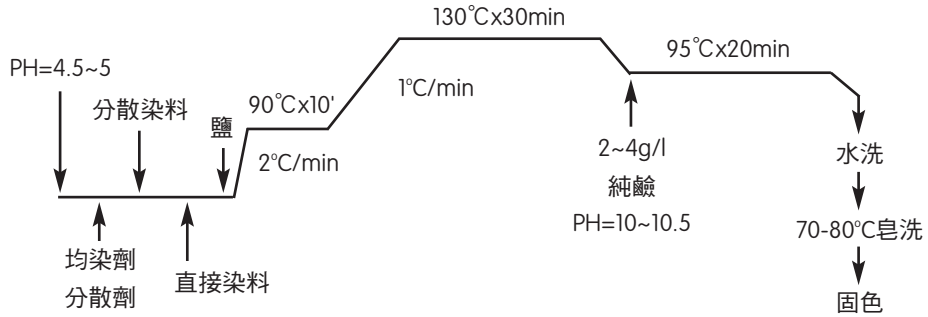
(6) 聚酯/纖維素纖維染色：

分散/直接染料一浴一段染法
(一般Everdirect染料皆可使用，但Everdirect Fast Black VSF 600除外)



(7) 聚酯/纖維素纖維染色：

分散/直接染料一浴二段染法（Everdirect Fast Black VSF 600）



附註：若分散染料對於鹽類過於敏感，則可將直接染料及鹽類於後段95℃時再加入。

3-4. 連續染色法：

直接染料之連續染色法主要為Pad-steam法：

(1) 壓吸過程：

壓吸過程必須以下列方式執行以避免“頭尾異色”：

(a) 使用小型染液槽提高染液交換率

(b) 染色物在染液槽的浸漬時間要保持一致

(c) 於低溫下進行壓吸

染料的溶解度是壓吸段中一項要因。對極淺色而言並無太大問題。但對染極深色則必須篩選高溶解度之染料。在pad-steam過程中為了提高染料的溶解度，通常會加入100 g/l的尿素以提高染料溶解度。

(2) 蒸處過程 (pad-steam)：

蒸處條件建議為102-105℃飽和蒸氣施染3-5分鐘。

染色物在經蒸處後再經一道30 g/l的鹽浴處理，可有效提高染色牢度與促進固著。

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染料選擇

1. 三原色

(A) 淺色系

Everdirect Yellow RL	Everdirect Supra Blue BGL *
Everdirect Supra Red BWS	Everdirect Supra Blue 4BL H/C*
Everdirect Supra Grey CGL	Everdirect Supra Blue GDL

(B) 中深色系

Everdirect Yellow RL	Everdirect Supra Blue BRR
Everdirect Supra Red BWS	Everdirect Supra Blue BGL*
Everdirect Supra Rubine RL	Everdirect Supra Blue 4BL H/C*
Everdirect Brown GTL	Everdirect Supra Blue BRL*
Everdirect Supra Blue GDL	Everdirect Supra Blue BRN*

2. 猩紅色

Everdirect Light Scarlet F2G	Everdirect Yellow RL
Everdirect Light Rose FR	

3. 翠綠色

Everdirect Supra Yellow PG	Everdirect Blue FFRL
Everdirect Turq. Blue FBL	Everdirect Turq. Blue GRL

4. 粉紅及紫色

Everdirect Light Rose FR	Everdirect Blue FFRL
輔助色：	
Everdirect Yellow RL	

5. 寶藍色

Everdirect Supra Blue BRL*	Everdirect Supra Blue BRN*
Everdirect Blue FFRL	Everdirect Light Rose FR
	Everdirect Supra Blue BRR

6. 黑色

Everdirect Fast Black VSF 600	Everdirect Fast Black B160
Everdirect Fast Black B300	Everdirect Black HRN
輔助色：	
Everdirect Yellow RL	Everdirect Supra Rubine BL
Everdirect Supra Red BWS	

備註：* 代表該品項於2014/12月停產

技術手冊內容

(1) Everdirect染料染色特性

1. S.D.C. 分類
2. 依染色吸盡溫度分類
3. 移染性
4. 溶解度
5. 多種纖維布之沾染試驗
6. 金屬離子影響性
7. 色恆性
8. 拔染性
9. 固色劑色變性
10. 熱變性

(2) Everdirect 染料牢度特性

1. 耐水洗牢度
2. 耐水牢度
3. 耐氯水牢度
4. 耐汗牢度
5. 耐日光牢度
6. 濕日光牢度
7. 耐汗日光牢度
8. 耐摩擦牢度
9. 氣體變褪色牢度

(3) 染色曲線

1. 染深性
2. PH依存性
3. 溫度依存性
4. 鹽依存性

Ⅰ.染色特性：

1. S.D.C. 分類：(Society of Dyers & Colorants)
   A 類：均染型
   B 類：鹽控型
   C 類：溫控型

A 類 (均染型)： 此類直接染料具極佳之移染性及均染性。染色時並無特殊注意事項，然而濕牢度不佳。

B 類 (鹽控型)： 此類染料非均染型，而必須以適當染法才能讓染料均勻上染於纖維。一旦不均染就很難修色。緩慢的加鹽有助於染料均勻上色及染深。

C 類 (溫控型)： 此類染料具有極強的親和力，單以緩慢的加鹽是無法控制其均染性。為了避免不均染，除了加鹽必須小量加入外，昇溫更必須緩慢。

2. 依染色吸盡溫度分類
   染色狀況：

材質	：	未絲光棉
染色深度	：	1% o.w.f.
浴比	：	1：10
鹽用量	：	10 g/l
溫度 & 時間	：	50°C x 60分鐘
		70°C x 60分鐘
		90°C x 60分鐘

   歸類：
   H型： 高溫型
   此類染料在較高溫下有較高的吸盡率，它們通常具有下列特性：
   ※ 染色上染速率慢，遮蓋性較差，但濕牢度佳。
   ※ 連染較低之頭尾異色機率。
   ※ 染棉及化纖混紡品比較容易控制色光，因為冷卻時回染量少。

M型：中溫型

染色溫度對此類染料無特別之影響性，它們通常具有下列特性：

※ 無須嚴格控制染色溫度，此類染料染色再現性佳。

※ Jigger 染色鮮少發生左右異色問題。

L 型： 低溫型

此類染料染料最高染色溫度65~75°C，染90°C時力度會減少。

因此此類染料染色必須小心控制染色溫度。此類染料的特點：

※ 移染性佳，均染性佳。此類染料染色初期瞬染性高，且相較其他染料濕牢度較差。

※ 注意連染時頭尾異色。

※ 染色冷卻過程的回染現象會造成色相差。

3. 移染性

染色狀況：

材質	：未絲光棉
染色深度	：1% o.w.f.
浴比	：1：10
鹽用量	：10 g/l
溫度 & 時間	：95~100°C x 30分鐘

移染試驗：

材質	：已染色布及未染棉布
染色深度	：1% o.w.f.
浴比	：1：10
鹽用量	：10 g/l
溫度 & 時間	：95~100°C x 30分鐘

分類：

移染試驗後以灰色標評估染色布及未染色布上之顏色差異。

4. 溶解度

將固定量之染料溶於90~95℃之熱水中，冷卻至80℃再以濾紙過濾。

過濾後依濾紙上之殘留物多寡判斷之。

5. 多種纖維染著試驗

染色條件：

材質	： 多種纖維布
染色深度	： 2 % o.w.f.
浴比	： 1：10
鹽用量	： 10 g/l
溫度 & 時間	： 95~100℃ x 30分鐘

評估： 依污染灰色標評估沾染程度

6. 金屬離子影響性

染色條件：

材質	： 未絲光棉
染色深度	： 1% o.w.f.
浴比	： 1：10
鹽用量	： 10 g/l
金屬離子	： Fe^{2+}離子 20 ppm
	Cu^{2+}離子 20 ppm
溫度 & 時間	： 95~100℃ x 30分鐘

評估：依色變灰色標評估不加金屬離子之染色布及加金屬離子染色布之間之色差。

7. 色恆性

以標準光源做基準，將在鎢絲燈和超效能日光燈（TL-84）下的演色性分為若干類表示之。

A：用AATCC變退色灰色標比較，相當於5－3-4級的，可作為微變色。
B：用AATCC變退色灰色標比較，相當於3-4－2-3級的，可作為相當程度變色。
C：用AATCC變退色灰色標比較，相當於2-3－1級的，可作為顯著變色。
（ ）內的記號為變色的傾向。

記號說明：

Y：黃色調　R：紅色調　B：藍色調　G：綠色調
P：紫色調　C：鮮明　D：暗調

8. 拔染性

拔染性由於底色的濃度和拔白處方的不同而效果不同。試驗用的染色棉織物之濃度是以2% o.w.f. 浸染染色的。

(1) 拔染糊處方：

組成/方法	中性	鹼性
拔染劑 (Decrolin)	100	---
拔染劑 (Rongalit C)	---	200
純鹼	---	10
吸濕劑	X	X
糊劑	Y	Y
全量	1000g	1000g

(2) 工程：印花→ 60°C烘乾 → 102°C x 5~8分鐘

9. 固色變色性

固色條件：

被染物　　：棉織物 (2% o.w.f.)

浴比　　：1：10

固色劑　　：聚醯胺型固色劑 2g/l

溫度 x 時間：60℃ x 20 分鐘

評估：依據灰色標判定織物固色及未固色之差異

10.熱變色性

染色條件：

被染物　　：未絲光棉

染色濃度　：1% o.w.f. (黑色染料 2% o.w.f.)

鹽用量　　：10g/l

浴比　　：1：10

溫度 x 時間：95~100℃ x 30 分鐘

評估：烘乾後立即以灰色標評估其色變。另外將測試樣置於室溫下2小時後再以色變灰色標評估色變。

測試：先將染色織物壓吸率控制至100%，然後將染色織物放置在 130℃/266℉ 熱風烘乾機中乾燥1分鐘。

II.牢度特性：

1. 牢度特性是以ISO、AATCC、JIS 和 CNS 為標準測試方法。
2. 牢度特性試驗是一般色以 1% o.w.f.，及黑色2% o.w.f.為染色深度。
3. 日光牢度特性是由指定的幾個標準深度。
4. 顏色的變化由以下的英文字縮寫表示：
   G：綠色調　Y：黃色調　R：紅色調　B：藍色調　D：暗調

(1) 耐水洗牢度：
ISO 105-C10 方法 B
AATCC-61- II A

(2) 耐水牢度：
ISO 105-E01

(3) 耐氯水牢度：
ISO 105-E03

(4) 耐汗牢度：
ISO 105-E04

(5)日光牢度：
ISO 105-B02
AATCC 16E

(6) 濕日光：
(M&S C9A)

(7) 汗日光牢度：
(ISO 105-B07)

(8) 摩擦牢度：
ISO 105-X12

(9) 氣體變褪色牢度：
ISO 105-G01

以上試驗除了日光牢度外，皆須經固色處理後再進行牢度試驗。

III.染色曲線

(1) 染深性（未絲光棉）

染色條件：

染料濃度　　：0.01、0.05、0.1、0.5、1、1.5、2、3、4、5、6% o.w.f.

浴比　　：1：10

鹽用量　　：3~10g/l

溫度 x 時間　　：98°C x 40 分鐘

(2) PH依存性（未絲光棉）

染色條件：

染料濃度　　：1% o.w.f.

浴比　　：1：10

染色之 PH　　：4、5、7、8、9、10、11

溫度 x 時間　　：98°C x 40分鐘

(3) 溫度依存性（未絲光棉）

染色條件：

染料濃度　　：2% o.w.f.

鹽用量　　：10g/l

浴比　　：1：10

染色之溫度　　：60、70、80、90、100°C

時間　　：40分鐘

(4) 鹽依存性（未絲光棉）

染色條件：

染料濃度　　：2% o.w.f.

鹽之用量 (芒硝)　：3、5、10*、15、20、30g/l

浴比　　：1：10

溫度 x 時間　　：98°C x 40分鐘

臺灣永光化學工業股份有限公司

Everlight Chemical Industrial Corporation

Everdirect

English Version

Everlight Chemical

Better Chemistry Better Life

Brand Promise

Through the passionate pursuit of progress and innovation, Everlight Chemical contributes to the successes of our customers. Our work helps create a better community and ensure a sustainable future. We believe that the intelligent application of chemistry can help make the world a better place.





Brand Values

Everlight Chemical believes that the intelligent application of chemistry can help make the world a better place. Everlight Chemical has earned recognition by customers and stakeholders for the following brand values:

Reliability

Everlight Chemical's brand trustworthiness is built upon a core value of integrity, determination to deliver on promises, stable product quality, and attentive service.

Client-Centric

Customer Caring with Value Creation is Everlight Chmical's Quality Policy. Our mission is to help clients achieve success through our high-quality products, innovative solutions, and technical support.

Innovation

Focusing on our core competency-Chemistry, Everlight Chemical highly values R&D that leads to breakthroughs and progress in developing chemical products that benefit humanity.

Welfare

Contribution to the welfare of society is high among Everlight Chemical's priorities. We hope our work can contribute to a brighter future for all of mankind.

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1. INTRODUCTION :

Everlight Chemical Industrial Corporation markets a wide range of high quality dircet dyes called "Everdirect dyes". These dyes have excellent fastness specifications.

Everdirect dyes are applied to all type of cellulosic fibers, such as cotton, viscose rayon and linen etc. Meanwhile, these dyes can achieve highly reliable wet fastness on viscose rayon.

Everdirect dyes are divided into four groups as follows :
Everdirect ———————— has excellent light fastness property.
Everdirect supra ——— has excellent light and wet fastness specifications.
Everdirect Light ——— has brilliant shade compared with reactive dyes.
Everdirect Cupro —— need an aftertreatment with copper-containing fixing agent to achieve excellent wet fastness specifications.

2. LIST OF EVERDIRECT DYES :

1. Everdirect Supra Yellow PG
2. Everdirect Yellow RL
3. Everdirect Supra Orange 2GL
4. Everdirect Light Scarlet F2G
5. Everdirect Supra Red BWS
6. Everdirect Light Rose FR
7. Everdirect Supra Rubine BL
8. Everdirect Brown GTL
9. Everdirect Turq. Blue FBL
10. Everdirect Turq. Blue GRL
11. Everdirect Blue FFRL
12. Everdirect Supra Blue BGL
13. Everdirect Supra Blue 4BL H/C
14. Everdirect Supra Blue BRL
15. Everdirect Supra Blue BRN
16. Everdirect Supra Blue GDL
17. Everdirect Supra Blue BRR
18. Everdirect Supra Grey CGL
19. Everdirect Black HRN
20. Everditrct Fast Black B160
21. Everdirect Fast Black VSF 600
22. Everdirect Black ANBN H/C
23. Everdirect Fast Black B300

3. GENERAL DYEING METHODS OF EVERDIRECT DYES :

Everdirect dyes are used to dye all type of cellulosic fibers and its blends by exhaustion, padding and printing methods. These dyes are ideal selection for exhaustion dyeing of cellulosic fibers.

3-1. Pretreatment of substrate :

The quality of pretreated substrate prior to dyeings will affcet the result of dyeing. Oil and sizing agent remained on the surface of the substrate should be removed as they may cause uneven dyeing. Chemical residues of the pretreatment should be throughly removed on account that it can cause substrate damage in the dyeing process. Therefore, the pretreatment process such as desizing, scouring, bleaching, mercerizing etc., should be carried out as careful as possible.

3-2. Exhaustion dyeing method :

Prepare a dyeing bath containing neutral salt (Glauber's salt or common salt), and put the substrate into the bath at about 30-50°C. Heat up the bath slowly to around boiling and maintain dyeing temperature for 30-60 min. After dyeing rinse the dyeing substrate with over-flow water to avoid excessive bleeding of the dyes. Finally, wet fastness can be improved by a fixing treatment in fresh bath with fixing agent.

(1) Dissolving procedure of dyes :

1. Prepare a dissoliving tank as large as possible.
2. Paste the weighed dyes with a small amount of cold water.
3. Add hot water and stir for a period of time.
4. Ensure the complete dissolution, add the solution into the dyebath through a filter cloth or screen.

(2) Recommended amount of neutral salt :

The necessary amount of neutral salt will depend on the substrate, dyes, depth of shade and liquor ratio. The recommended amount of neutral salt is shown as belows :

Dyeing depth % o.w.f.	
extremely pale shade (less than 0.05%以下)	0 - 2 g/l
pale shade (0.05-1.0%)	2 - 5 g/l
medium shade (1.0 - 4.0%)	5 - 10 g/l
heavy shade (more then 4.0%)	20 g/l

The table list is based on liquor ratio 1:10

* For Everdirect Fast Black B-160, Fast Black B300 and Fast Black VSF 600, it is necessary to add 2-4 g/l soda ash into the dyebath in order to give a PH value of 10.0-10.5.

(3) Fixing treatment :

Dyed substrate is immersed into a fixing bath under the following conditions :

Fixing agent	: 0/5-2.0 g/l
Liquor ratio	: 1:10
Temp. & Time	: 60°C x 20-30 min.

(4) Correction of uneven dyeing :

When uneven dyeings are found immediately after dyeing process, they can be corrected by the migration or partial stripping of the dyes on the substrate in the following conditions :

Soda ash	: 3-5 g/l
Non-ionic surfactant	: 2-3 g/l
Sodium pyrophosphate	: 1 g/l
Temp. & Time	: 100°C x 30 min.

A higher liquor ratio gives a better effect on the correction.

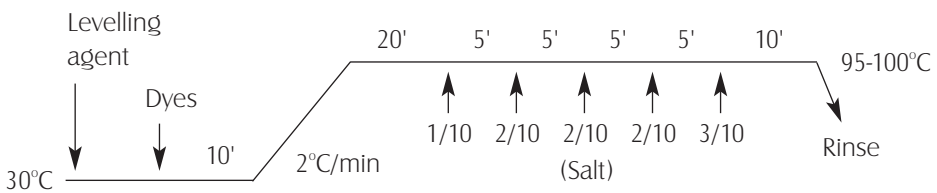
(5) Stripping of dyes :

Sodium hydrosulfite : 3-5 g/l
Caustic soda. (flake) : 2 g/l
Temp. & Time : 80-90°C x 30 min

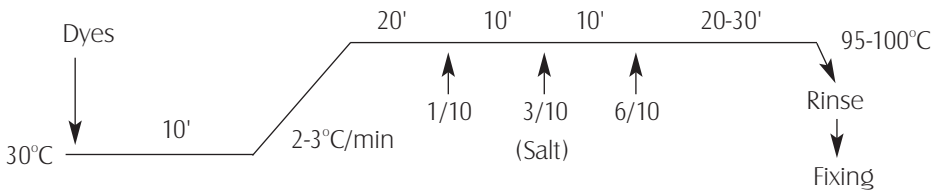
It is very difficult to strip the dyes evenly and throughly, even severe stripping gives an undesired influence on the handling touch of the dyeing substrate. In such cases, it is better to re-dye the substrate a darker shade after a moderate stripping.

3-3. Exhaust dyeing methods of Everdirect dyes :

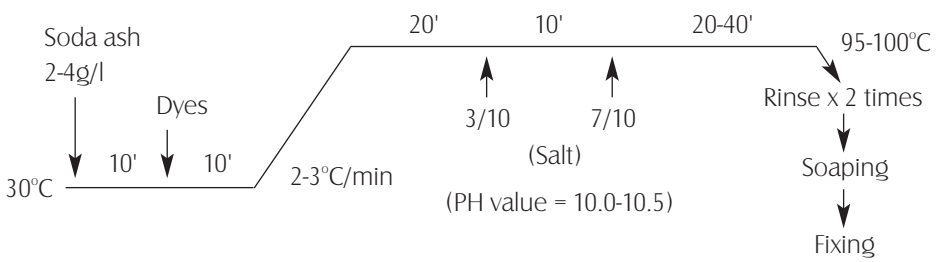
(1) Pale shades :



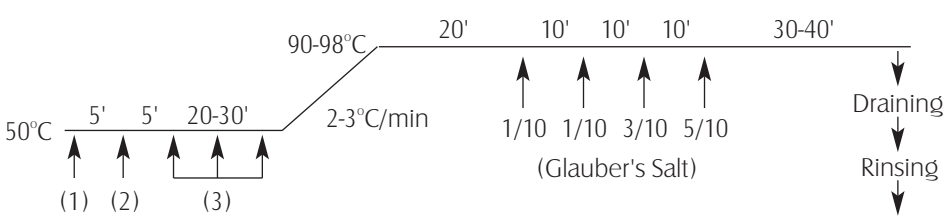
(2) Medium to heavy shades :



(3) Special dyeing method of Everdirect Fast Black B-160, Fast Black B300 & Fast Black VSF 600 :



(4) Special dyeing method of Everdirect Supra Blue FBL & GRL Dyeing curve :



(1) Penetrating agent : 1%
(2) Levelling agent : 1%
(3) Dyes : 2-3 portionwise addition.

(5) Dyes & Salt at start :

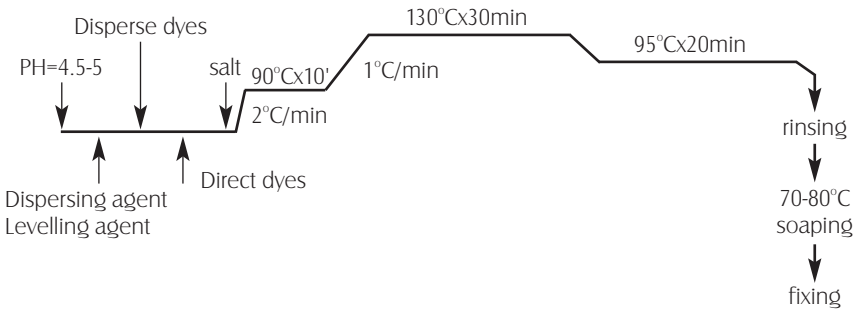


REMARKS :

1. Turquoise Blue FBL & GRL are also suitable for T/C one bath dyeing.
2. Liquor ratio should be controlled between 1:10 to 1:20.
3. Glauber's salt is prefered, instead of common salt.
4. After dyeing, the rinsing process is very important. Insufficient rinsing will cause unlevelness. Besides, after fixing, the fabric must be dried immediately.

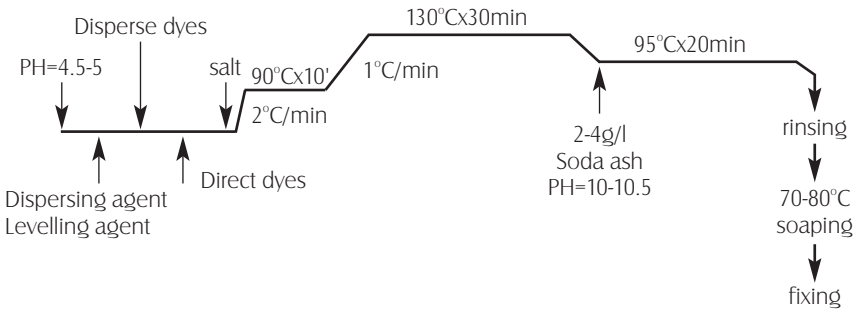
(6) Polyester/Cotton blends dyeing :

Disperse/Direct dyes at one bath one step method :
(this method is widely recommended for Everdirect dyes, except Everdirect Fast Black VSF 600)



(7) Polyester/Cotton blends dyeing :

Disperse/Direct dyes at one bath two steps method :
(Especially for Everdirect Fast Black VSF 600)



3-4. Continuous dyeing method :

The continuous dyeing method is divided into two stages. The first stage is: Substrate to be padded with a dyes solution. The second stage is: dyes on the substrate to be developed. The continuous dyeing method is mainly based on pad-steam process.

(1) Padding stage : (The first stage)

To minimize "Tailing" effect, the procedure should be carried out in the padding stage as follows:

(a) Use a small padding trough to increase the exchange of fresh dyes solution. (pick-up: 70~100%)
(b) Shorten the immersion time of the substrate in padding solution.
(c) Pad the substrate as lower pick-up as possible.
(d) Pad the substrate at low temperature. (In the case of heavy shade, the trough is often heated up to 50-60°C, in order to maintain good dissolution of dye.)

The solubility of the dyes is the another concern in the padding stage. There will be no trouble in the pale shade. However, selection of proper dyes with high solubility are necessary in heavy shade. An addition of about 100 g/l of urea will improve the solubility of the dyes, in order to increase the final uptake of the dyes in pad-steam process.

In the padding stage, an addition of 1-2 g/l of sodium alginate can get good effect of migration property.

(2) Steaming stage : (The second stage of pad-steam process)

In the pad-steam process, the substrate exits from the padder and enters into the steamer. The steamer must be filled with saturated vapor in order to fix the substrate. Steaming stage is carried out at 102-105°C for 3-5 minutes.

In case that steamer is not large enough for the requested steaming time, the substrate should be run though another boiling salt bath 30 g/l after steaming.

Everlight Chemical

Dye Selection

1. Tertiary Shade :

(A) Pale shades

Everdirect Yellow RL	Everdirect Supra Blue BGL *
Everdirect Supra Red BWS	Everdirect Supra Blue 4BL H/C*
Everdirect Supra Grey CGL	Everdirect Supra Blue GDL

(B) Medium-deep shade

Everdirect Yellow RL	Everdirect Supra Blue BRR
Everdirect Supra Red BWS	Everdirect Supra Blue BGL*
Everdirect Supra Rubine RL	Everdirect Supra Blue 4BL H/C*
Everdirect Brown GTL	Everdirect Supra Blue BRL*
Everdirect Supra Blue GDL	Everdirect Supra Blue BRN*

2. Scarlet Red

Everdirect Light Scarlet F2G	Everdirect Yellow RL
Everdirect Light Rose FR	

3. Jade green & Aqua green

Everdirect Supra Yellow PG	Everdirect Blue FFRL
Everdirect Turq. Blue FBL	Everdirect Turq. Blue GRL

4. Pink/Violet

Everdirect Light Rose FR	Everdirect Blue FFRL
Everdirect Yellow RL	

5. Royal Blue

Everdirect Supra Blue BRL*	Everdirect Supra Blue BRN*
Everdirect Blue FFRL	Everdirect Light Rose FR
Everdirect Supra Blue BRR	

6. Black

Everdirect Fast Black VSF 600	Everdirect Fast Black B160
Everdirect Fast Black B300	Everdirect Black HRN

supplementary :

Everdirect Yellow RL	Everdirect Supra Rubine BL
Everdirect Supra Red BWS	

Note：The marked " * " products had been stop production since Dec. 2014

(Ⅰ) Dyeing properties of Everdirect dyes :

1. S.D.C. classification
2. Exhaustion classification by temperature
3. Migration property
4. Solubility
5. Multifiber cross-dyeing
6. Effect of metallic ions
7. Color Constancy
8. Dischargeability
9. Shade change by fixing agent
10. Shade change by heat

(Ⅱ) Fastness properties of Everdirect dyes :

1. Washing
2. Water
3. Chlorinated water
4. Perspiration
5. Light
6. Wet- Light
7. Perspiration-Light
8. Rubbing
9. Colorfastness to Nitrogen Oxides

(Ⅲ) Dyeing curves :

1. Build-up
2. PH dependency
3. Temperature dependency
4. Salt dependency

(Ⅰ) Dyeing properties :

1. S.D.C. Classifications : (Society of Dyers and Colorants)

Class A : Self-levelling
Class B : Salt-controllable
Class C : Temperature-controllable

Class A : These direct dyes have good migration property and levelness. They can be applied without special precaution. However, the wet fastness are poor.

Class B : These direct dyes are not self-levelling and must go onto the fibers in a levelling manner. Any unlevelness will be difficult to correct. A gradual addition of salt will give a uniform uptake and build-up of the dyes.

Class C : These direct dyes have strong affinity whose exhaustion can not be controlled adequately by the addition of salt alone. Once unlevelness – which is very difficult to correct– it is necessary to raise the temperature slowly while making small addition of salt.

2. Exhaustion classification by temperature

Dyeing conditions :

Material	: Unmercerized cotton
Dipth of shade	: 1% o.w.f.
Liquor ratio	: 1:10
Glauber's salt	: 10 g/l
Temp. & Time	: 50°C, 60 min.
	70°C, 60 min.
	90°C, 60 min.

Evaluation :

H type : High temperature type.

Dyes show higher exhaustion at higher dyeing temperature, and they generally have the following characteristics:

* Slower rate of dyeing, poor covering property, but good wet fastness.
* Lower tailing tendency in continuous dyeing.
* In synthetic fiber with cellulosic blends dyeing, it's easy to control the dyeing shade because of re-exhaustion occures rarely during the cooling period.

M type : Medium temperature type.

These dyes show no particular relation with the dyeing temperature, and they generally have the following characteristics :

* No need to control the dyeing temperature strictly, and these dyes have very good dyeing reproducibility.
* Ending effect rarely happens in jigger dyeing.

L type : Low temperature type.

These dyes show maximun exhaustion at 65-75°C and the strength reduction at 90°C. The dyeing temperature should be controlled precisely. They have the following characteristics :

* Good levelling property because of the excellent migration ability. These dyes have high strike at initial dyeing and have poor wet fastness than the other types of dyes.
* Beware of tailing in continuous dyeing.
* Shade change may occur by re-exhaustion during the cooling process.

3. Migration property

Dyeing conditions :

Material	: Unmercerized cotton
Depth of shade	: 1% o.w.f.
Liquor ratio	: 1:10
Glauber's salt	: 10 g/l
Temp & Time	: 95-100°C x 30 min.

Migration test :

Material	: dyed and undyed cotton fabric
Liquor ratio	: 1:10
Glauber's salt	: 10g/l
Temp. & Time	: 95-100°C x 30 min.

Evaluation :

After the migration test, assess the contrast between the dyed and undyed cotton fabric by using the grey scale.

4. Solubility

A predetermined amount of dyes were dissolved in hot water (90-95°C), and then cooling down to 80°C, the dyes solution was filtered by filter paper. The solubility was measured on the presence or absence of residue on the filter paper.

5. Multifibers cross-dyeing

Dyeing conditions :

Material	: Multifibers
Depth of shade	: 2% o.w.f.
Glauber's salt	: 10g/l
Liquor ratio	: 1:10
Temp. & Time	: 95-100°C x 30 min

Evaluation :

The degree of cross-dyeing on multifiber was assessed by the grey scale.

6. Effect of metallic ions

Dyeing conditions :

Material	: Unmercerized cotton
Depth of shade	: 1% o.w.f.
Glauber's salt	: 10g/l
Liquor ratio	: 1:10
Metallic ions	: Fe^{++} (20 ppm as Fe^{++})
	Cu^{++} (20 pm as Cu^{++})
Temp. & Time	: 95-100°C x 30 min.

Evaluition :

The shade difference and yield between the dyed samples and without adding metallic ion were assessed by the grey scale.

7. Metamerism

The shades of dyed meteials under tungsten filament lamp and daylight fluorescent lamp were compared with that under standard light source, and the degree of shade difference was measured and classified as follows :

A : Slight shade difference in shade perceived corresponding to a rating range, 5 to 3-4, of the grey scale for assessing the color change.

B : Considerable shade difference perceived coresponding to a rating range, 3-4 to 2-3, of the same grey scale.

C : Much shade difference perceived corresponding to a rating range, 2-3 to 1, of the same grey scale.

Symbols in parentheses mean the hue tendency of shade difference.

Y : Yellower R : Redder B : Bluer G : Greener

P : Purpler C : Birighter D : Duller

8. Dischargeability

Dischargeability varies with the depth of ground dyeing and recipe of white discharge. The discharge test was made on cotton fabrics dyed at 2% o.w.f.

(1) Recipe of discharge paste :

Composition/Method	Neutral	Alkaline
Decrolin	100	—
Rongalic C	—	200
Soda ash	—	10
Glycerine	30	30
Thickener	770	760
Total	1000 parts	1000 parts

(2) Procedure :

After printing, the fabrics were dried at 60°C, steamed at 102°C for 5 to 8 minutes, followed by rinsing and soaping.

9. Shade change by Fixing agent

Fixing conditions:

Material	: dyed cotton fabric (2% o.w.f.)
Liquor ratio	: 1:10
Fixing agent	: Polyamine fixing agent 2g/l
Temp. & Time	: 60°C x 20 min

Evaluation :

The shade difference between the fixed sample and without adding fixing agent were assessed by the grey scale.

10. Shade change by Heat

Dyeing conditions:

Material	: Unmercerized cotton
Depth of shade	: 1% o.w.f. (Black Dyes 2% o.w.f.)
Glauber's salt	: 10 g/l
Liquor ratio	: 1:10
Temp. & Time	: 95-100°C x 30 min

Indication :

Immediately after drying, the shade change of the dyeings is assessed, and again after 2 hours re-condition at room temperature, and compared to the one at the room temperatrue by the grey scade for assessing shade change.

Evaluation :

After squeezing the dyed fabric by controlled pick-up 100%, then the dyed fabric is kept in a hot-air dryer at 130°C/266°F for 1 min.

(II) Fastness properties :

1. The fastness properties were assessed according to the standards of the ISO, AATCC, Fastness test results are listed.
2. The fastness properties were tested with dyeing depth of 1.0% o.w.f. for normal dyes, 2% o.w.f. for black dyes.
3. The light fastrness property is stated for several standard depth.
4. The hue tendency of a shade change is indicated by the following abbreviations: G=greener Y=yellower R=redder B=bluer D= duller

(1) Washing :
(ISO 105 C10 B)
AATCC-61-IIA

(2) Water :
(ISO 105-E01)

(3) Chlorinated Water :
(ISO 105-E03)

(4) Perspiration :
(ISO 105-E04)

(5) Light :
(ISO 105-B02)

(6) Wet-Light :
(M&S C9A)

(7) Persiration-Light :
(ISO 105-B07)

(8) Rubbing :
(ISO 105-X12)

(9) Colorfastness to Nitrogen Oxides :
(ISO 105-G01)

above tests were carried out under fixing treatment, except light fastness.

(Ⅲ) Dyeing Curves :

(1) Build-up

Dyeing conditions :

Material	: Unmercerized cotton
Depth of shade	: 0.01、0.05、0.1、0.5、1、1.5、2、3、4、5、6% o.w.f.
Liquor ratio	: 1:10
Glauber's salt	: 3-10 g/l
Temperature & Time	: 98°C x40min.

(2) PH dependency

Dyeing conditions :

Material	: Unmercerized cotton
Depth of shade	: 1% o.w.f.
Liquor ratio	: 1:10
pH of dyebath	: 4、5、7、8、9、10、11
Temperature & Time	: 98°C x 40min.

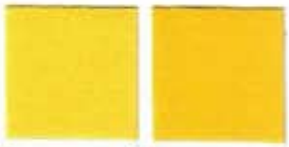
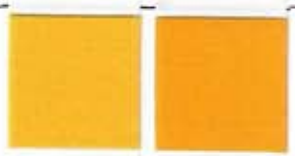
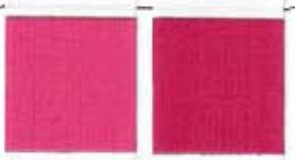
(3) Temperature dependency

Material	: Unmercerizecd cotton
Depth of shade	: 2.0% o.w.f.
Glauber's salt	: 10 g/l
Liquor ratio	: 1:10
Dyeing temperature	: 60、70、80、90、100*°C
Dyeing time	: 40min.

(4) Salt dependency

Material	: Unmercerized cotton
Depth of shade	: 2.0% o.w.f.
Glauber's salt	: 3, 5, 10*, 15, 20, 30, g/l
Liquor ratio	: 1:10
Temperature & Time	: 98°C x 40min.

Everdirect Dyes

頁 page	染料 dyes	1% 3%	染料特性	Characteristics
1	Supra Yellow PG		1. 鮮艷、高色值且帶綠味的黃色 2. 均染性及移染性佳 3. 日光及耐氯水牢度佳	1. Brilliant & greenish yellow dyes with high color-yield. 2. Excellent levelness and migration property. 3. Excellent lightfastness and chlorinated water fastness.
3	Yellow RL		1. 高色值偏紅味的黃色 2. 與R-BWS, B-4BL為經濟性三原色 3. 染色適用性廣 4. 耐光牢度佳，是黃色染料中最佳	1. Reddish yellow of high color-yield. 2. Compatible with R-BWS, B-4BL as economical trichromatic. 3. Versatile Application. 4. Highest lightfastness among yellow colorants.
5	Supra Orange 2GL		1. 高色值的橙色 2. 耐氯水及日光牢度佳，有輕微光變性 3. 可拔白的染料	1. Essential orange dyes of high color-yield. 2. High chlorinated water fastness and lightfastness, slight photochromism. 3. dischargeable dyes.
7	Light Scarlet F2G		1. 極鮮豔的大紅色，與Rose FR可染鮮紅色 2. 耐高溫安定性佳，適合T/C一浴染色	1. Very brilliant scarlet, compatible with Rose FR as brilliant red. 2. Excellent high temperature stability and suitable for T/C one bath dyeing.
9	Supra Red BWS		1. 與Y-RL, B-4BL為經濟性三原色 2. 高溫安定性佳 3. 中等的濕牢度	1. Economical red component as trichromatic with Y-RL, B-4BL. 2. Excellent high temperature stability. 3. Moderate wet fastness
11	Light Rose FR		1. 鮮艷色且帶藍味之紅色，均染性佳 2. 耐高溫安定性佳及耐雙氧水漂特性佳，可做T/C一浴染及漂染一浴 3. 濕牢度佳	1. Brilliant bluish red, excellent levelness. 2. Excellent high temperature stability and peroxide bleaching property, suitable for T/C one bath dyeing and scouring / bleaching & dyeing in one bath. 3.Excellent wet fastness.
13	Supra Rubine BL		1. 優異的濕牢度，適合做高堅牢度染色用三原色的紅色 2. 可搭配O-2GL, B-BRN作為深三原色 3. 日光牢度佳	1. Recommended as red component for high wet fastness. 2. Compatible with O-2GL, B-BRN as deep shade trichromatic. 3. Exceptional lightfastness.
15	Brown GTL		1. 中性色系的棕色 2. 適合作深色系之套色用 3. 日光牢度佳 4. 適合漂染一浴	1. Neutral brown dyes. 2. Useful as a shading component for dark shade. 3. Excellent lightfastness. 4. Suitable for scouring/bleaching and dyeing in one bath.
17	Turq. Blue FBL		1. 鮮豔之翠藍色 2. 搭配Y-PG可染翠綠色 3. 均染性極佳	1. Brilliant turquoise dyes. 2. Compatible with Y-PG for jade green shade dyeing. 3. Excellent levelness performance.
19	Turq. Blue GRL		1. 帶綠味之翠藍色 2. 搭配Y-PG可染翠綠色 3. 均染性極佳	1. Brilliant greenish turquoise dyes. 2. Compatible with Y-PG for jade green shade dyeing. 3. Excellent levelness performance.
21	Blue FFRL		1. 均染性佳且日光牢度佳的寶藍色 2. 搭配Rose FR可染鮮豔色之紫色 3. 注意溶解度	1. Excellent levelness and high lightfastness as a royal blue dyes. 2. Compatible with Rose FR for brilliant violet shade dyeing. 3. Pay attention to low solubility.

備註：* 代表該品項於2014/12月停產

Note：The marked " * " products had been stop production since Dec. 2014

Everdirect Dyes

頁 page	染料 dyes	1% 3%	染料特性	Characteristics
23	* Supra Blue BGL		1. 良好的均染性 2. 日光牢度佳 3. 適合練漂一浴，T/C一浴 4. 可與Y-RL、R-BWS、Ru-BL搭配	1. Unique levelness performance 2. Excellent lightfastness 3. Suitable for souring/bleaching & dyeing in one bath. 4. Compatible with Y- RL、R-BWS、Ru-BL.
25	* Supra Blue 4BL H/C		1. 濕牢度佳，可作為高堅牢度用三原色之藍色 2. 可與Y-RL, R-BWS, Ru-BL搭配 3. 高溫安定性佳，適合T/C一浴染色	1. Excellent wet fastness as the blue component of basic trichromatic. 2. Compatible with Y-RL, R-BWS, Ru.-BL. 3. Suitable for T/C one bath dyeing as excellent high temperature stability.
27	* Supra Blue BRL		1. 染深性佳及帶紅味之藏青色 2. 優異的濕牢度 3. 耐高溫染色安定性佳，可適合做T/C一浴染色	1. Excellent build-up and reddish navy dyes. 2. Excellent wet fastness. 3. Excellent high temperature stability, suitable for T/C one bath dyeing.
29	* Supra Blue BRN		1. 染深性佳及帶紅味之藏青色 2. 優異的濕牢度 3. 耐高溫染色安定性佳，可適合做T/C一浴染色	1. Excellent build-up and reddish navy dyes. 2. Excellent wet fastness. 3. Excellent high temperature stability, suitable for T/C one bath dyeing.
31	Supra Blue GDL		1. 染深性佳及帶綠味之藏青色 2. 優異的濕牢度 3. 耐高溫染色安定性佳，可適合做T/C一浴染色	1. Excellent build-up and greenish navy dyes. 2. Excellent wet fastness. 3. Excellent high temperature stability, suitable for T/C one bath dyeing.
33	Supra Blue BRR		1. 染深性佳及帶紅味之藏青色 2. 優異的濕牢度 3. 耐高溫染色安定性佳，可適合做T/C一浴染色	1. Excellent build-up and reddish navy dyes. 2. Excellent wet fastness. 3. Excellent high temperature stability, suitable for T/C one bath dyeing.
35	Supra Grey CGL		1. 極佳之均染性及移染性 2. 可用於淺色單色染色或配色 3. 日光牢度佳 4. 耐高溫安定性佳，可用於T/C一浴染色	1. Excellent levelness and migration property. 2. Suitable for single dyeing or shading. 3. Excellent lightfastness. 4. Excellent hight temperature stability, suitable for T/C one bath dyeing.
37	Black HRN		1. 經濟染深之黑色 2. 在鹼性浴下可染深，故可用於T/C一浴二段染色	1. Economical and excellent build-up. 2. Suitable for T/C dyeing in one stage two step method.
39	Fast Black B160		1. 經濟染深之黑色 2. 在鹼性浴下可染深 3. 可用於漂染一浴	1. Economical and excellent build-up. 2. Dyeable at alkaline bath. 3. Suitable for bleaching and dyeing in one bath.
41	Fast Black VSF 600		1. 染深性極佳之黑色 2. 在鹼性浴下可染深，不建議在酸性下染色 3. 可用於漂染一浴 4. 軋染時注意溶解度 5. 可用於T/C一浴二段染色	1. Economical and excellent build-up. 2. Dyeable at alkaline bath. 3. Suitable for bleaching and dyeing in one bath. 4. Noticeable solubility at jigger dyeing. 5. Suitable for T/C dyeing in one stage two step method.
43	Black ANBN H/C		1. 染深性佳之黑色 2. 在酸性浴下具優異的再現性	1. Excellent build-up. 2. Excellent reproducibility under acid bath.
45	Fast Black B300		1. 染深性極佳之黑色 2. 在鹼性浴下可染深 3. 可用於漂染一浴	1. Economical and excellent build-up 2. Dyeable at alkaline bath. 3. Suitable for bleaching and dyeing in one bath.

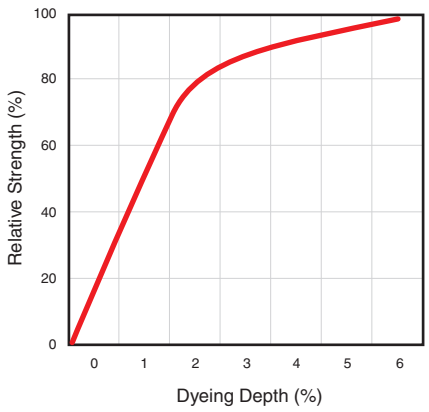
Everlight Chemical

Everdirect Supra Yellow PG

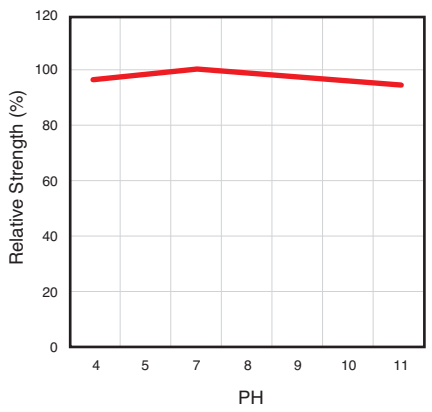
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		M
Migration Property 移染性		3-4
Solubility g/l 溶解度	80°C Water	50
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	1
	Polyester	4-5
	Acrylic	4-5
	Wool	2
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	1-2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4-5
Shade change by heat 熱變色性	0	4-5G
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Yellow PG

Everlight Chemical

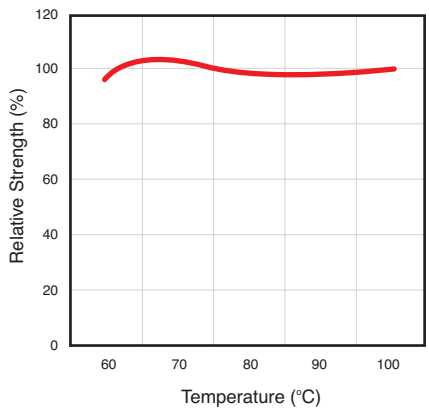
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4	5	1-2	5	5	5	5
	ISO C10 B		1.0%	4-5	5	1-2	4-5	5	5	5
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	4-5						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

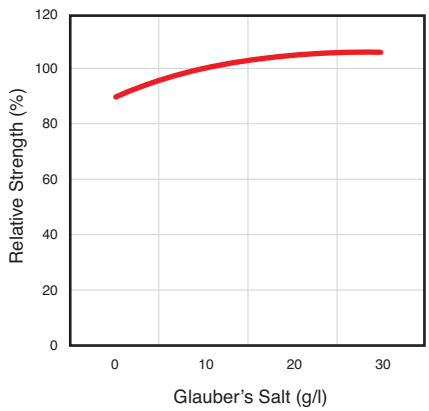
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	5-6
	0.1 %	5-6		Alkali		5R
	0.5 %	6	Wet-Light 耐濕日光		0.2 %	5-6
	1.0 %	>6			2.0 %	6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

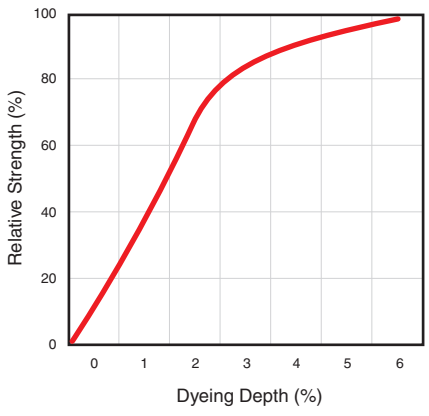


Everdirect Yellow RL

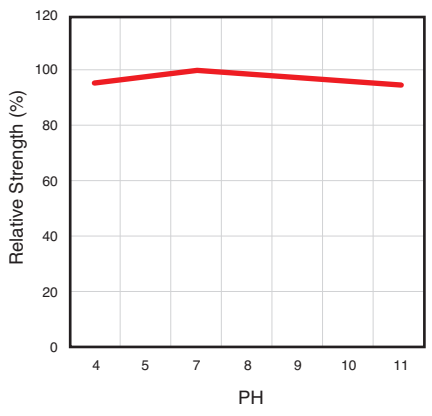
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		M
Migration Property 移染性		3
Solubility g/l 溶解度	80°C Water	80
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	2
	Polyester	4-5
	Acrylic	4-5
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Discharge -ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	1-2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4-5
Shade change by heat 熱變色性	0	4
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Yellow RL

Everlight Chemical

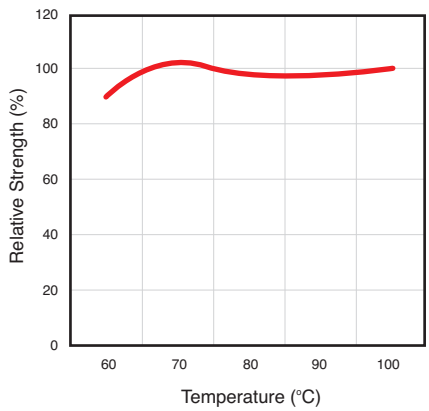
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	2-3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	2	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	4-5						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

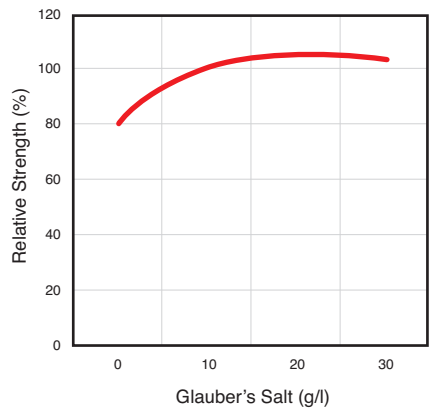
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	>6
	0.1 %	6		Alkali		>6
	0.5 %	6	Wet-Light 耐濕日光		0.2 %	6
	1.0 %	>6			2.0 %	6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

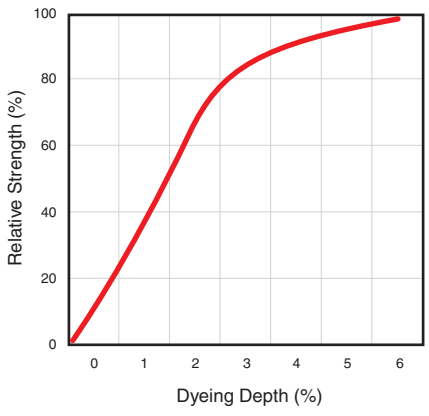


Everdirect Supra Orange 2GL

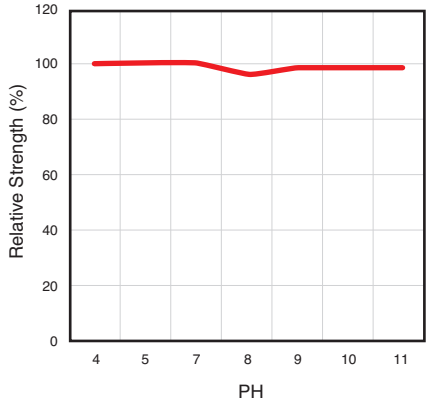
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		L
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	100
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	1
	Polyester	4-5
	Acrylic	4
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	Y
	TL 84 Lamp	Y
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	1-2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4-5R
Shade change by heat 熱變色性	0	4YG
	2hr	4-5YG

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Orange 2GL

Everlight Chemical

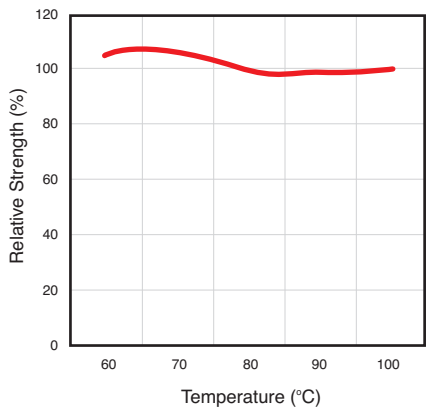
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	2-3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	2	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	4-5						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

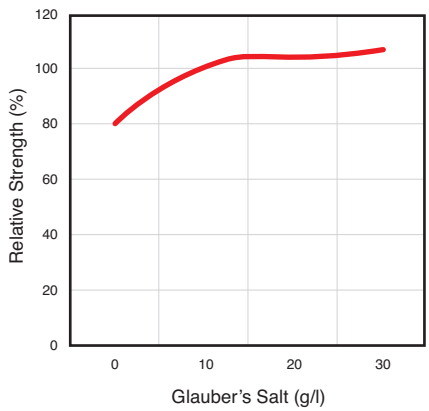
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	>6	Perspiration Light 耐汗日光	Acid	2.0 %	>6
	0.1 %	>6		Alkali		5-6R
	0.5 %	>6	Wet-Light 耐濕日光		0.2 %	6
	1.0 %	>6			2.0 %	6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

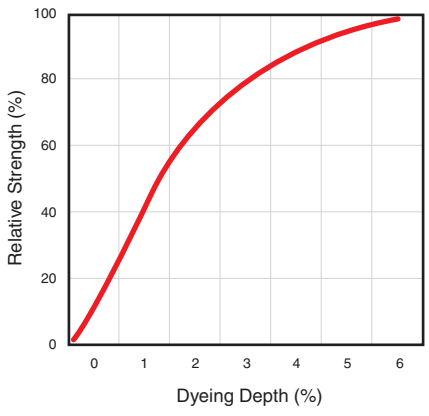


Everdirect Light Scarlet F2G

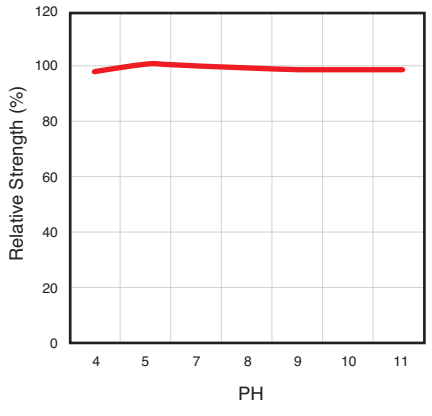
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	20
Cross-Dyeing 多種纖維沾染	Acetate	3-4
	Cotton	1
	Nylon	2
	Polyester	4-5
	Acrylic	4-5
	Wool	1-2
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Discharge -ability 拔染性	Neutral 中性	2
	Alkaline 鹼性	2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3Y
Shade change by heat 熱變色性	0	3-4B
	2hr	4-5Y

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Light Scarlet F2G

Everlight Chemical

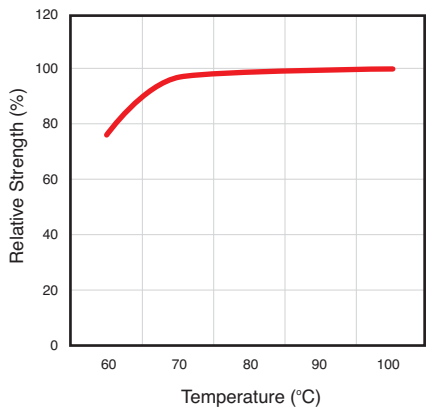
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	2-3	5	5	5	5
	ISO C10 B		1.0%	4R	5	3	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4R	5	5	5	5	5	5
		Alkali 鹼		4R	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	3R						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

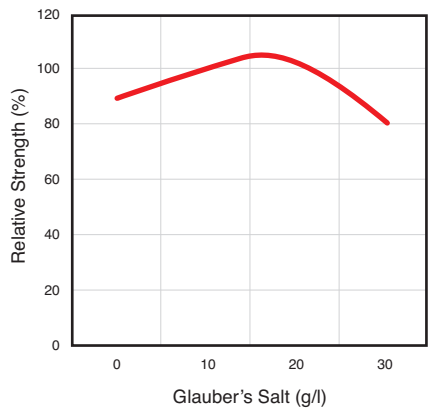
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	3-4
	0.1 %	1-2		Alkali		3-4
	0.5 %	2-3	Wet-Light 耐濕日光		0.2 %	2Y
	1.0 %	3			2.0 %	3-4
	2.0 %	3-4				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

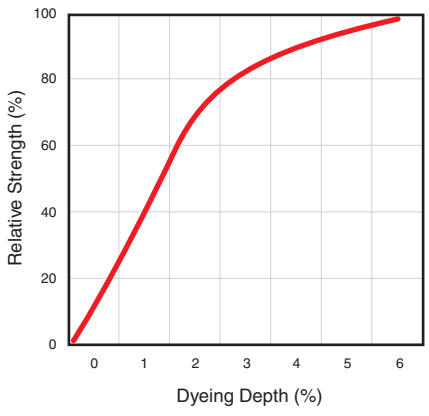


Everdirect Supra Red BWS

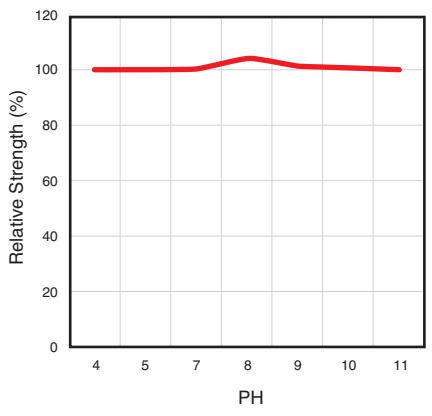
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		3
Solubility g/l 溶解度	80°C Water	100
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	1
	Polyester	4-5
	Acrylic	4-5
	Wool	2
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	B
	TL 84 Lamp	Y
Discharge -ability 拔染性	Neutral 中性	2
	Alkaline 鹼性	2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4-5Y
Shade change by heat 熱變色性	0	4
	2hr	4-5Y

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Red BWS

Everlight Chemical

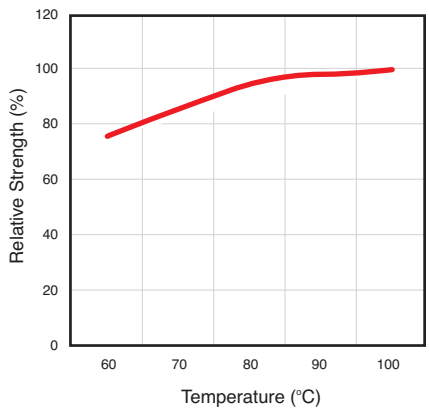
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	1-2	5	5	5	5
	ISO C10 B		1.0%	4-5	5	2-3	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4R	5	4	4-5	5	5	5
		Alkali 鹼		4R	5	3-4	4-5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	4Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

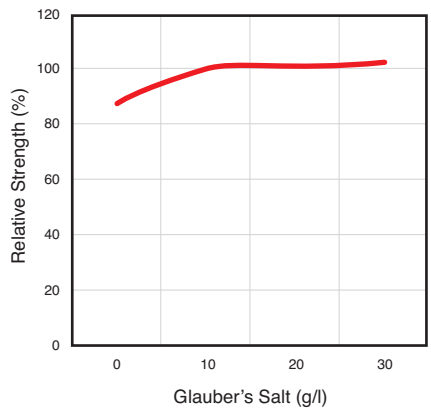
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	3-4	Perspiration Light 耐汗日光	Acid	2.0 %	4-5
	0.1 %	4		Alkali		4-5
	0.5 %	4	Wet-Light 耐濕日光		0.2 %	4
	1.0 %	4-5			2.0 %	4-5
	2.0 %	4-5				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

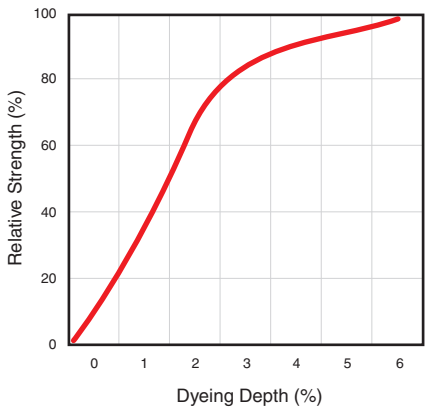


Everdirect Light Rose FR

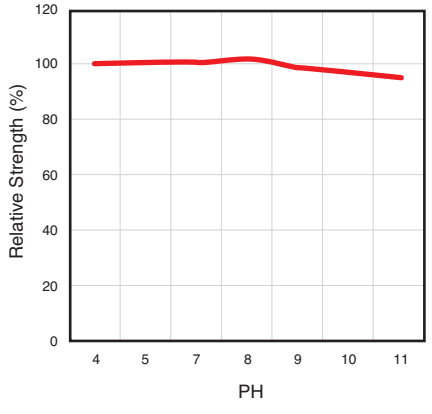
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	100
Cross-Dyeing 多種纖維沾染	Acetate	4
	Cotton	1
	Nylon	1-2
	Polyester	4-5
	Acrylic	4-5
	Wool	1-2
Metallic ions 金屬離子	Fe^{2+} 鐵	3-4
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	B
	TL 84 Lamp	Y
Discharge -ability 拔染性	Neutral 中性	2
	Alkaline 鹼性	2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4B
Shade change by heat 熱變色性	0	4
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Light Rose FR

Everlight Chemical

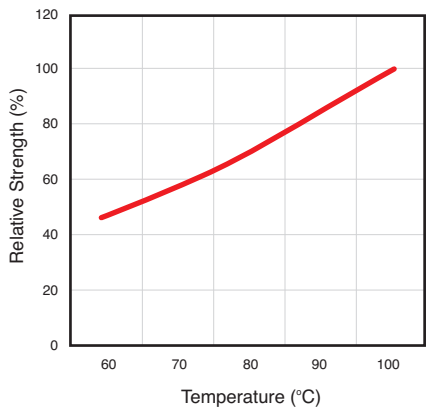
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	4	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4R	5	5	5	5	5	5
		Alkali 鹼		4R	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	1Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

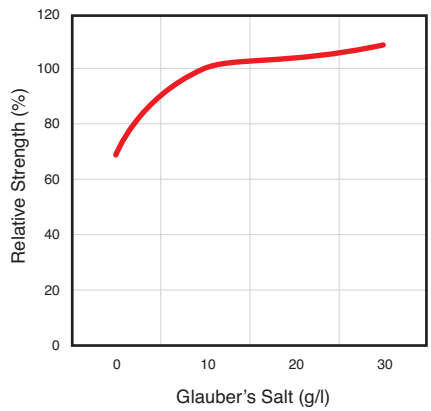
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	3-4B
	0.1 %	1-2		Alkali		3-4B
	0.5 %	2-3	Wet-Light 耐濕日光		0.2 %	2
	1.0 %	3			2.0 %	3-4
	2.0 %	3-4				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

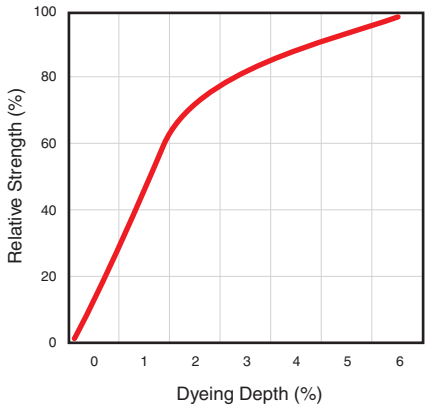


Everdirect Supra Rubine BL

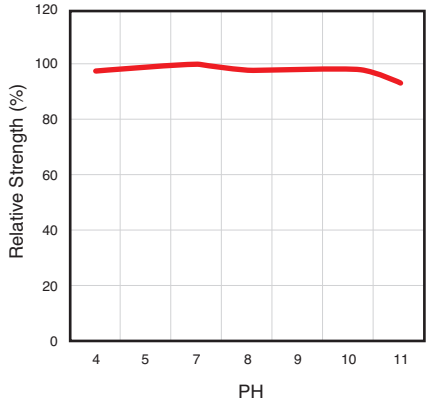
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		1-2
Solubility g/l 溶解度	80°C Water	60
Cross-Dyeing 多種纖維沾染	Acetate	5
	Cotton	1
	Nylon	2-3
	Polyester	4-5
	Acrylic	4-5
	Wool	3
Metallic ions 金屬離子	Fe^{2+} 鐵	4Y
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	B
	TL 84 Lamp	Y
Discharge -ability 拔染性	Neutral 中性	2
	Alkaline 鹼性	2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3-4B
Shade change by heat 熱變色性	0	4B
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Rubine BL

Everlight Chemical

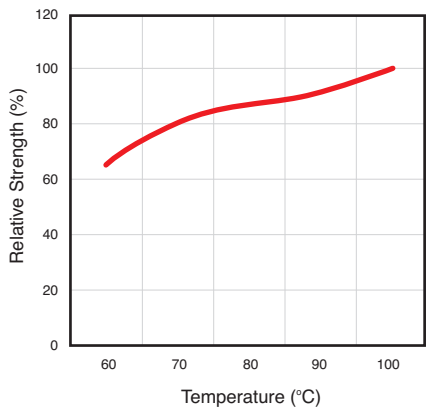
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	4-5	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	2Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

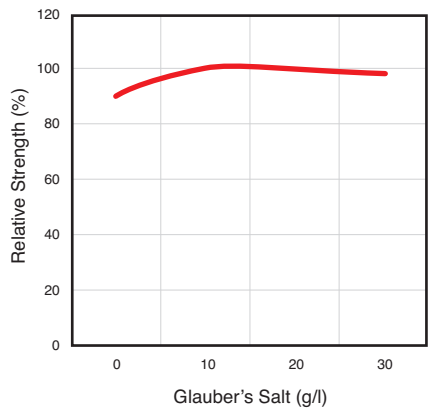
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	5-6
	0.1 %	6		Alkali		5-6Y
	0.5 %	>6	Wet-Light 耐濕日光		0.2 %	5-6
	1.0 %	>6			2.0 %	5-6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

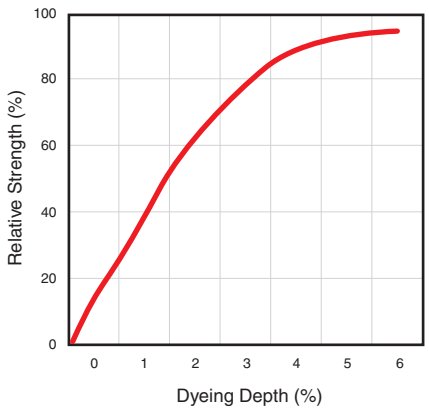


Everdirect Brown GTL

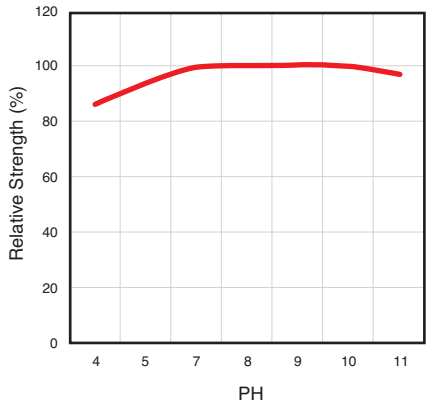
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		M
Migration Property 移染性		1-2
Solubility g/l 溶解度	80°C Water	70
Cross-Dyeing 多種纖維沾染	Acetate	3
	Cotton	1
	Nylon	1-2
	Polyester	4-5
	Acrylic	4
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4R
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Discharge -ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4
Shade change by heat 熱變色性	0	4
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Brown GTL

Everlight Chemical

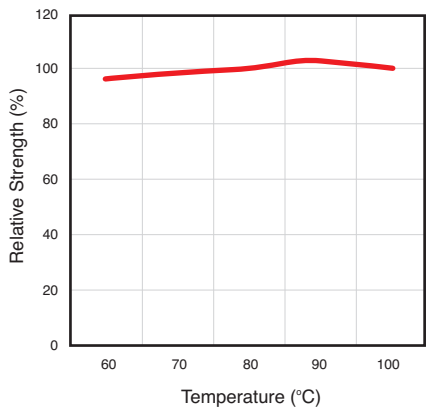
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3-4	5	5	5	5
	ISO C10 B		1.0%	4-5	5	3-4	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5R	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	3Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4-5						

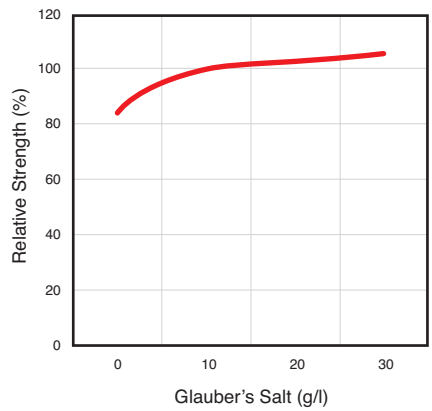
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	5-6
	0.1 %	5-6		Alkali		5-6
	0.5 %	5-6	Wet-Light 耐濕日光		0.2 %	5Y
	1.0 %	5-6			2.0 %	5-6
	2.0 %	5-6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

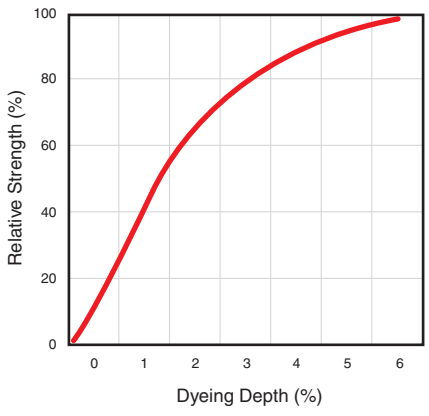


Everdirect Turq. Blue FBL

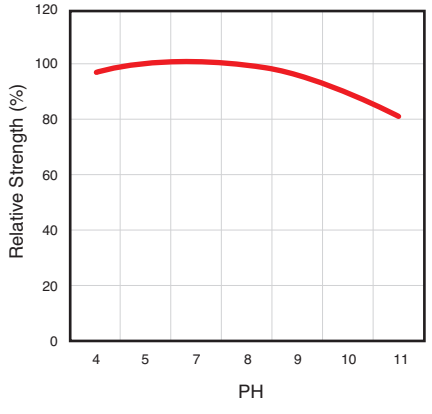
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		4
Solubility g/l 溶解度	80°C Water	60
Cross-Dyeing 多種纖維沾染	Acetate	4
	Cotton	1
	Nylon	1
	Polyester	4
	Acrylic	1-2
	Wool	1-2
Metallic ions 金屬離子	Fe^{2+} 鐵	3-4
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	2-3
	Alkaline 鹼性	2
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4Y
Shade change by heat 熱變色性	0	3-4YG
	2hr	4YG

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Turq. Blue FBL

Everlight Chemical

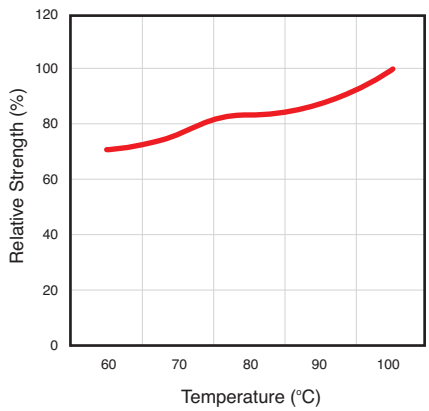
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	4	4-5	5	5	5
	ISO C10 B		1.0%	4-5	5	2-3	4	5	4	5
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	2Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4						

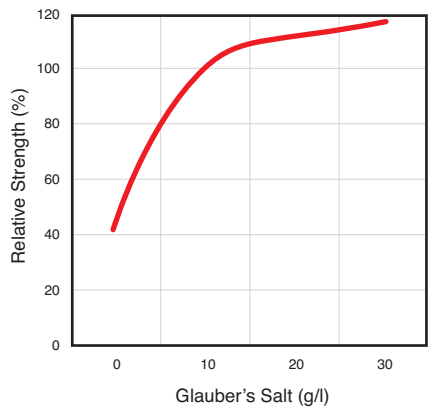
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	3-4Y
	0.1 %	5-6		Alkali		3-4Y
	0.5 %	5-6	Wet-Light 耐濕日光		0.2 %	5-6
	1.0 %	5-6			2.0 %	5-6
	2.0 %	5-6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

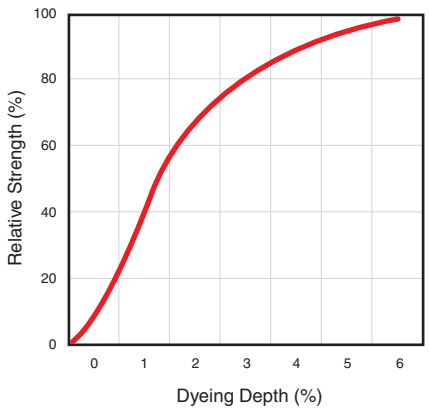


Everdirect Turq. Blue GRL

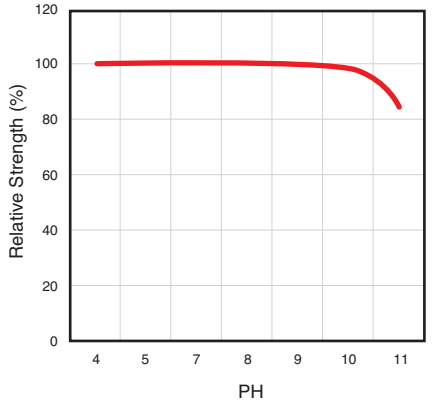
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		4
Solubility g/l 溶解度	80°C Water	60
Cross-Dyeing 多種纖維沾染	Acetate	4
	Cotton	1
	Nylon	1
	Polyester	4
	Acrylic	1-2
	Wool	1-2
Metallic ions 金屬離子	Fe^{2+} 鐵	3-4YD
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	2-3
	Alkaline 鹼性	2-3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3-4Y
Shade change by heat 熱變色性	0	3YG
	2hr	3-4YG

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Turq. Blue GRL

Everlight Chemical

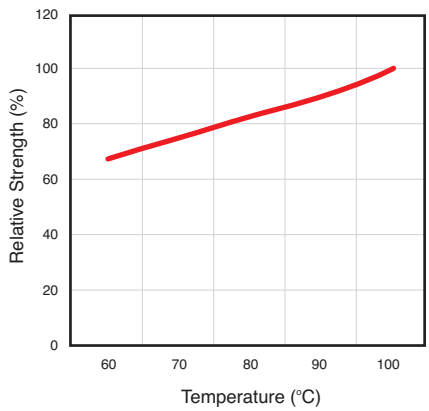
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	3-4	4-5	3	2	2-3	3	4-5
	ISO C10 B		1.0%	4	3-4	1	1	3	1	4
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	3Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4						

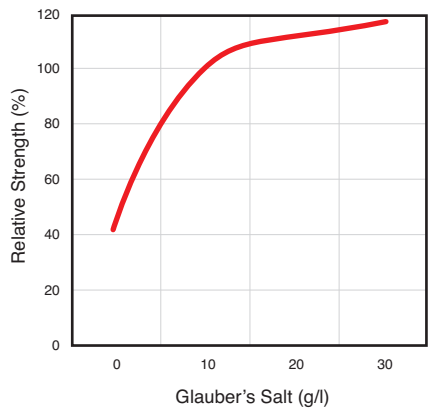
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	3-4Y
	0.1 %	5-6		Alkali		3-4Y
	0.5 %	5-6	Wet-Light 耐濕日光		0.2 %	5-6
	1.0 %	5-6			2.0 %	5-6
	2.0 %	5-6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

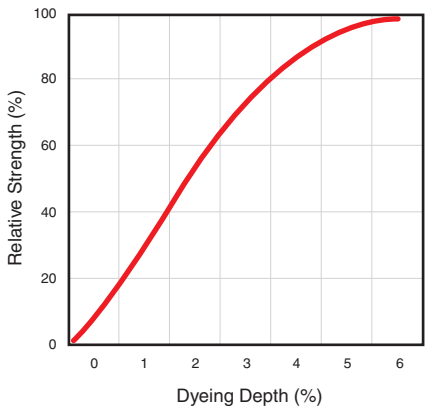


Everdirect Blue FFRL

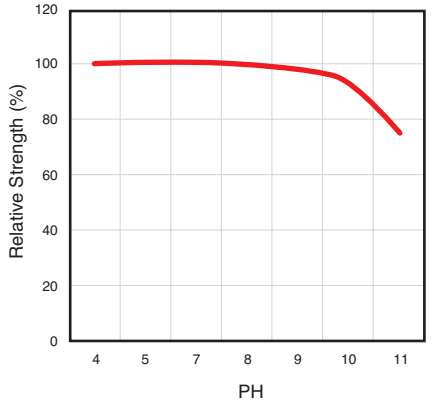
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		M
Migration Property 移染性		2-3
Solubility g/l 溶解度	80°C Water	10
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	2
	Polyester	3-4
	Acrylic	4-5
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	3
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	1
	Alkaline 鹼性	1
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3
Shade change by heat 熱變色性	0	3-4
	2hr	4Y

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Blue FFRL

Everlight Chemical

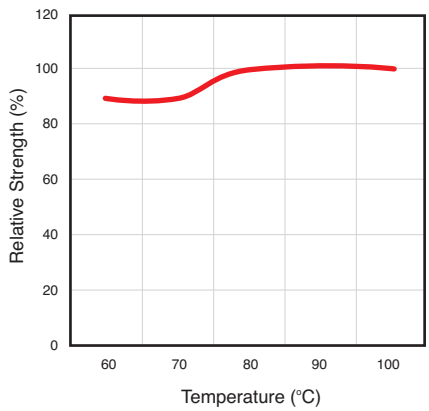
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	1	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	2R						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	4						

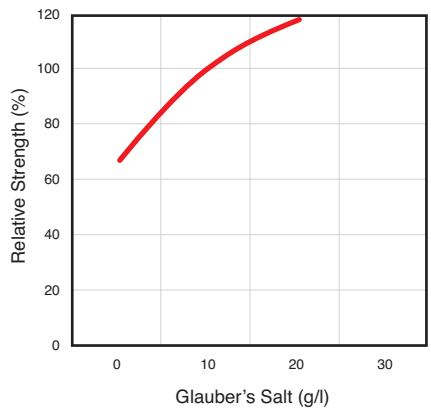
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	5-6
	0.1 %	6		Alkali		5-6
	0.5 %	6	Wet-Light 耐濕日光		0.2 %	5-6
	1.0 %	>6			2.0 %	5-6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

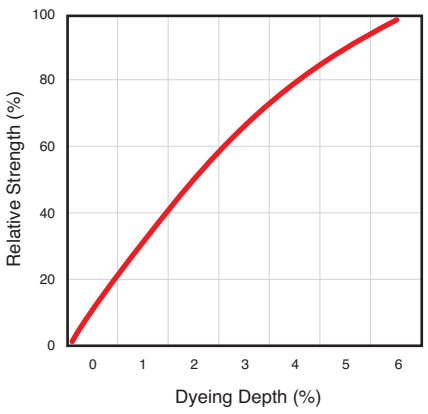


Everdirect Supra Blue BGL

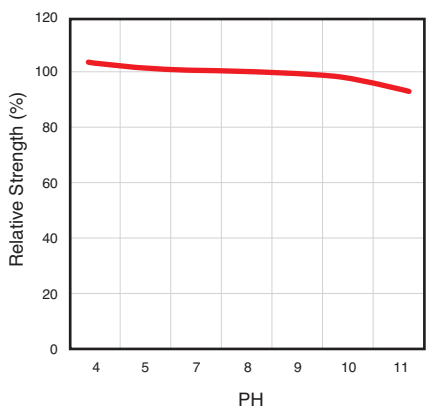
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2-3
Solubility g/l 溶解度	80°C Water	40
Cross-Dyeing 多種纖維沾染	Acetate	3-4
	Cotton	1
	Nylon	3
	Polyester	4
	Acrylic	3-4
	Wool	2-3
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	B
Discharge -ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	3-4
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3-4
Shade change by heat 熱變色性	0	3R
	2hr	4R

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Blue BGL

Everlight Chemical

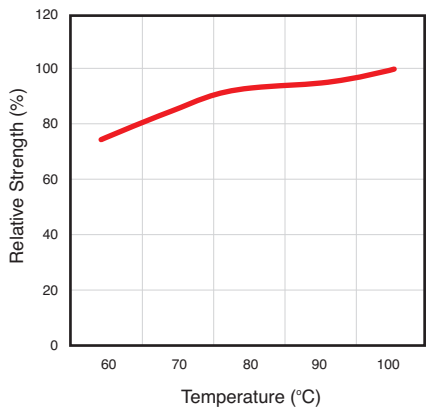
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	3-4	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	2						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3						

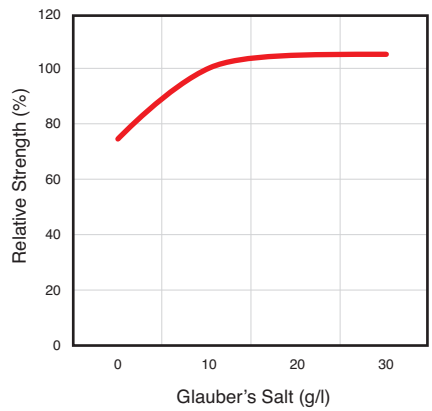
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	3	Perspiration Light 耐汗日光	Acid	2.0 %	5
	0.1 %	3-4		Alkali		4-5
	0.5 %	4	Wet-Light 耐濕日光		0.2 %	4-5
	1.0 %	5			2.0 %	4-5
	2.0 %	5-6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

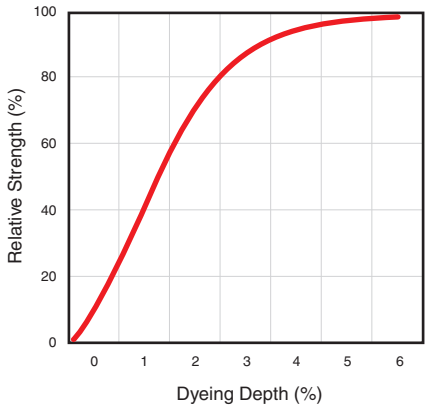


Everdirect Supra Blue 4BL H/C

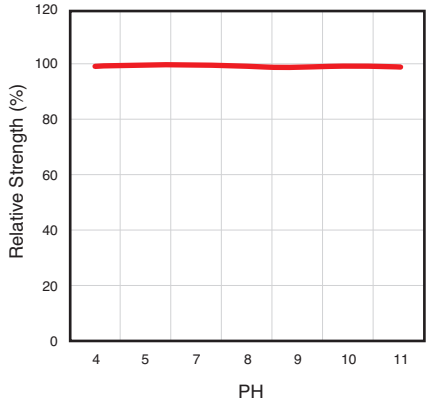
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		3
Solubility g/l 溶解度	80°C Water	50
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	2-3
	Polyester	4-5
	Acrylic	3
	Wool	2-3
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	2-3
	Alkaline 鹼性	2-3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4Y
Shade change by heat 熱變色性	0	4R
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Blue 4BL H/C

Everlight Chemical

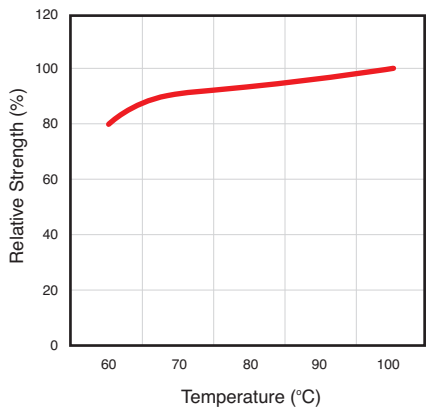
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	4	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	1-2R						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3-4						

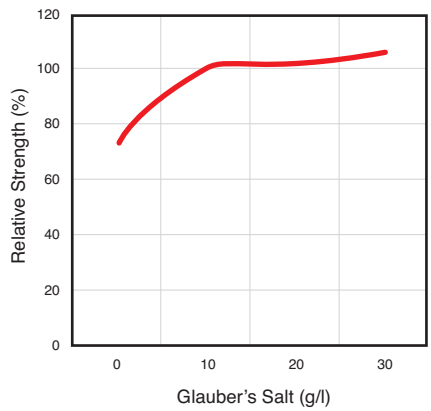
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	3	Perspiration Light 耐汗日光	Acid	2.0 %	5
	0.1 %	3-4		Alkali		5
	0.5 %	4	Wet-Light 耐濕日光		0.2 %	4R
	1.0 %	5			2.0 %	5
	2.0 %	5				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

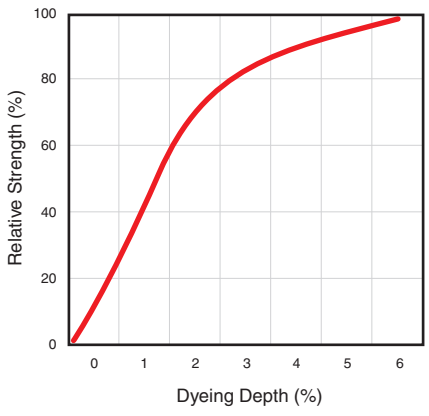


Everdirect Supra Blue BRL

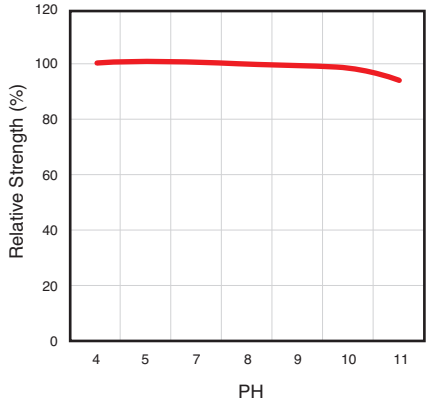
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	50
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	1-2
	Polyester	4-5
	Acrylic	3
	Wool	2
Metallic ions 金屬離子	Fe^{2+} 鐵	4
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4Y
Shade change by heat 熱變色性	0	3-4R
	2hr	5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Blue BRL

Everlight Chemical

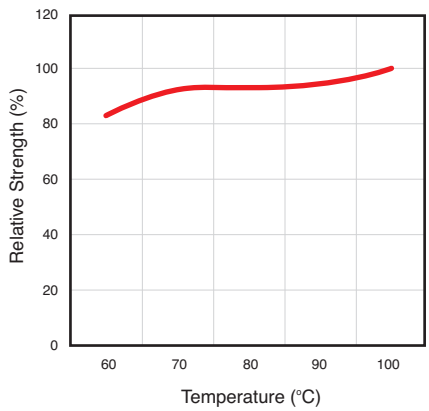
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4	5	4-5	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	1-2R						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3						

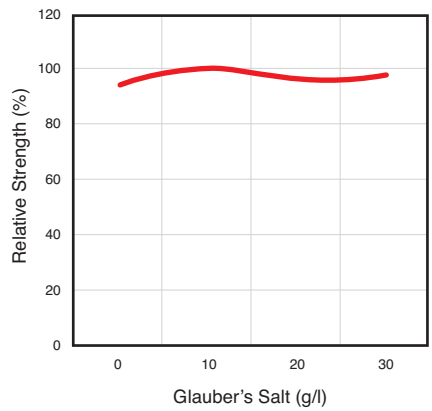
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	3	Perspiration Light 耐汗日光	Acid	2.0 %	5
	0.1 %	3		Alkali		5
	0.5 %	4	Wet-Light 耐濕日光		0.2 %	2R
	1.0 %	5			2.0 %	5
	2.0 %	5				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

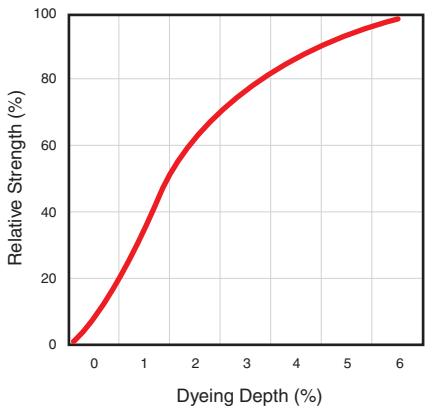


Everdirect Supra Blue BRN

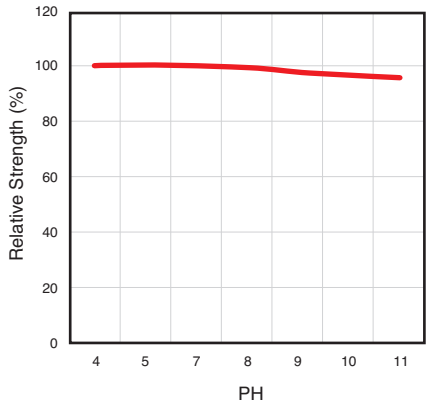
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	30
Cross-Dyeing 多種纖維沾染	Acetate	4
	Cotton	1
	Nylon	2-3
	Polyester	4-5
	Acrylic	2-3
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	2-3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3-4
Shade change by heat 熱變色性	0	4G
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Blue BRN

Everlight Chemical

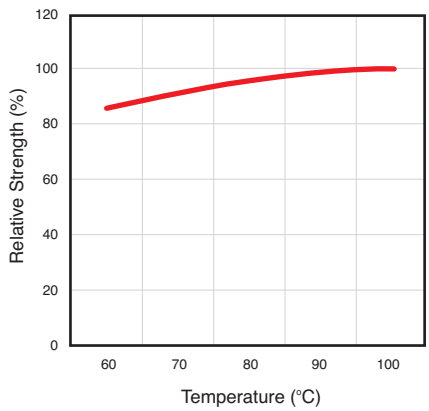
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	3	5	5	5	5
	ISO C10 B		1.0%	4-5	5	3-4	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		2-3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	1R						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3						

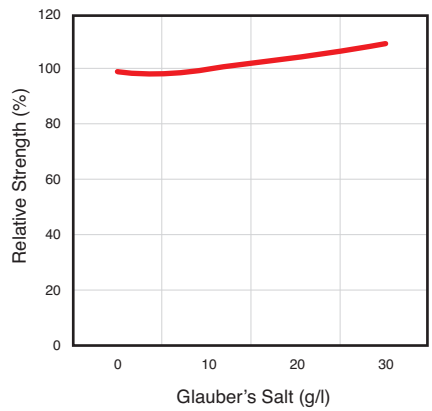
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	4	Perspiration Light 耐汗日光	Acid	2.0 %	5
	0.1 %	4-5		Alkali		5
	0.5 %	5	Wet-Light 耐濕日光		0.2 %	3-4
	1.0 %	5			2.0 %	5
	2.0 %	5				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

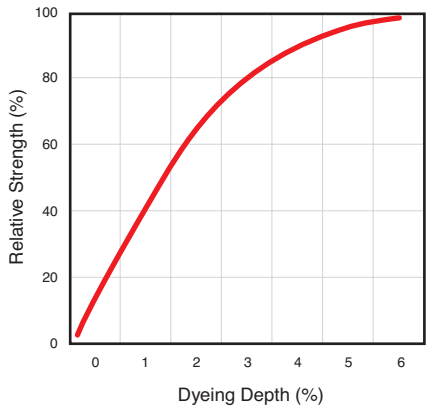


Everdirect Supra Blue GDL

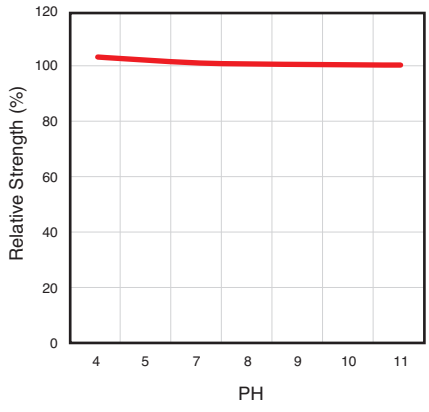
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		1-2
Solubility g/l 溶解度	80°C Water	30
Cross-Dyeing 多種纖維沾染	Acetate	4
	Cotton	1
	Nylon	3
	Polyester	4-5
	Acrylic	4
	Wool	2
Metallic ions 金屬離子	Fe^{2+} 鐵	5
	Cu^{2+} 銅	5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	B
Discharge -ability 拔染性	Neutral 中性	4
	Alkaline 鹼性	4
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3
Shade change by heat 熱變色性	0	4R
	2hr	4-5G

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Blue GDL

Everlight Chemical

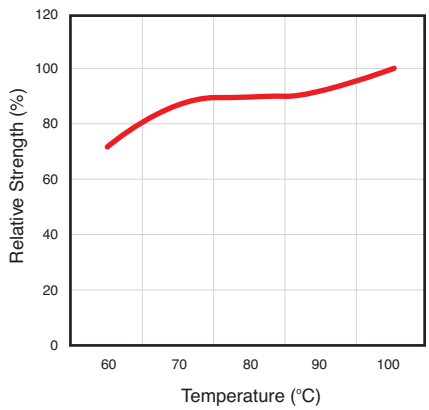
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	3-4	5	2	4-5	5	5	5
	ISO C10 B		1.0%	4	4-5	2	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	4-5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	4-5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	2						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3						

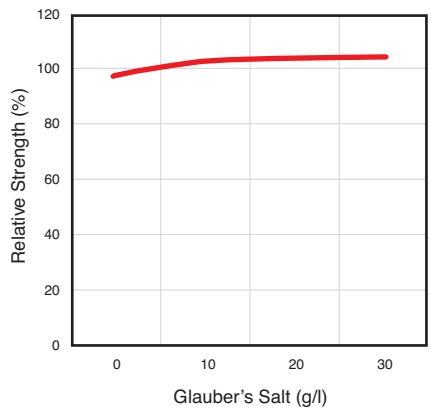
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	4	Perspiration Light 耐汗日光	Acid	1.0 %	6
	0.1 %	4-5		Alkali		5
	0.5 %	5	Wet-Light 耐濕日光		0.2 %	4-5
	1.0 %	5-6			2.0 %	5
	2.0 %	6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

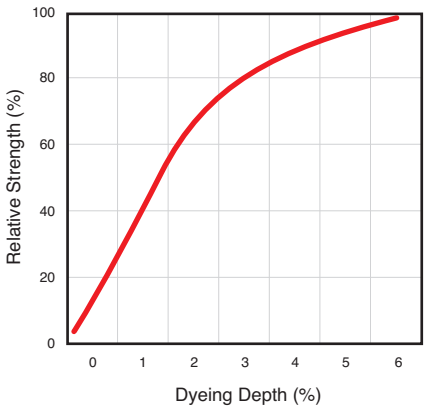


Everdirect Supra Blue BRR

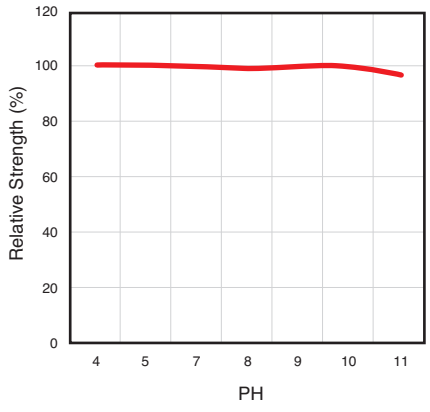
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2-3
Solubility g/l 溶解度	80°C Water	30
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	1
	Polyester	4-5
	Acrylic	4
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5G
	Cu^{2+} 銅	4-5R
Color constancy 色恆性	Tungsten Lamp	B
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	4
	Alkaline 鹼性	4
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	3-4
Shade change by heat 熱變色性	0	3-4
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Blue BRR

Everlight Chemical

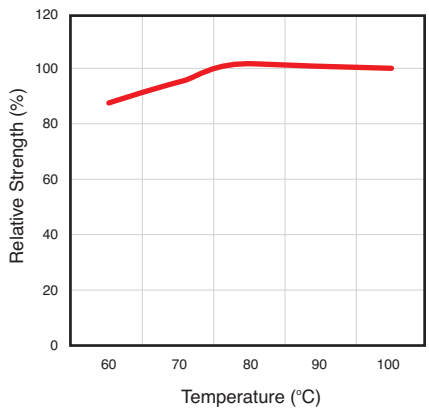
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	3-4	4-5	3	3-4	4-5	4-5	4-5
	ISO C10 B		1.0%	4	4-5	2	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	2-3						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3						

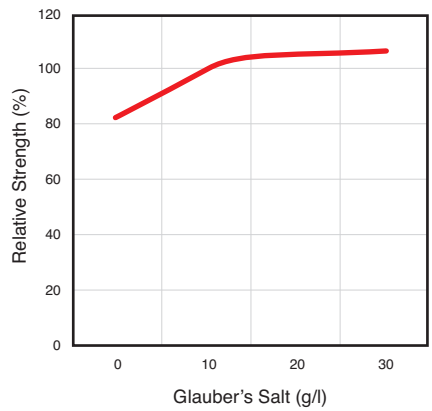
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	3	Perspiration Light 耐汗日光	Acid	2.0 %	5-6
	0.1 %	4		Alkali		5-6
	0.5 %	5	Wet-Light 耐濕日光		0.2 %	4-5
	1.0 %	6			2.0 %	6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

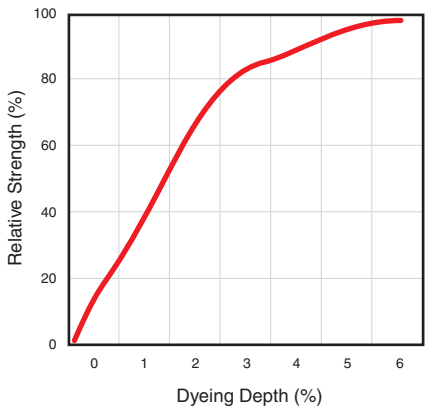


Everdirect Supra Grey CGL

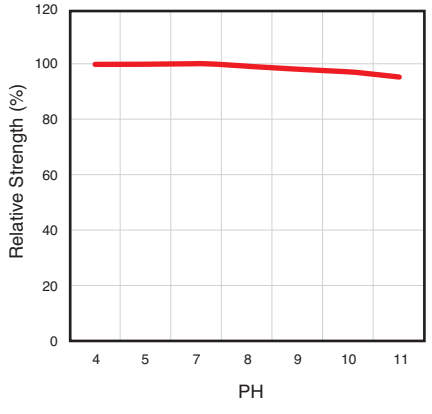
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2-3
Solubility g/l 溶解度	80°C Water	70
Cross-Dyeing 多種纖維沾染	Acetate	4-5
	Cotton	1
	Nylon	2
	Polyester	4-5
	Acrylic	3
	Wool	3
Metallic ions 金屬離子	Fe^{2+} 鐵	4
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4Y
Shade change by heat 熱變色性	0	4
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Supra Grey CGL

Everlight Chemical

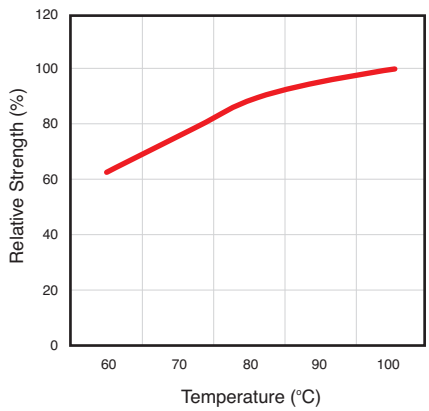
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		1.0%	4-5	5	4	5	5	5	5
	ISO C10 B		1.0%	3-4	5	5	5	5	5	5
Water 耐水	ISO 105-E01		1.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	1.0%	4-5R	5	5	5	5	5	5
		Alkali 鹼		4-5R	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	1.0%	5						
		Wet 溼		3-4						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			1.0%	3Y						
Colorfastness to Nitrogen Oxides 氣體變褪色			1.0%	3						

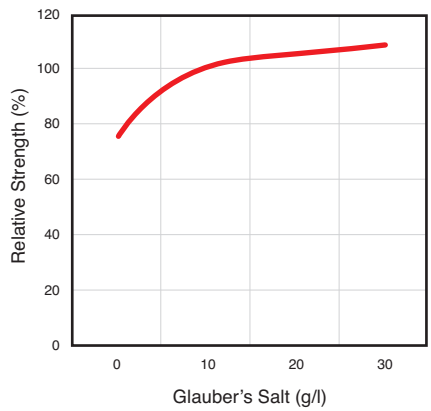
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	5-6	Perspiration Light 耐汗日光	Acid	2.0 %	5-6
	0.1 %	5-6		Alkali		5-6
	0.5 %	5-6	Wet-Light 耐濕日光		0.2 %	5-6
	1.0 %	6			2.0 %	6
	2.0 %	>6				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

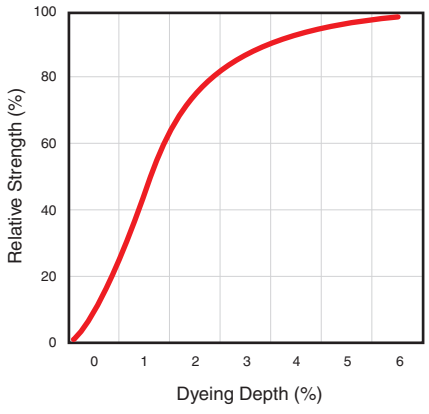


Everdirect Black HRN

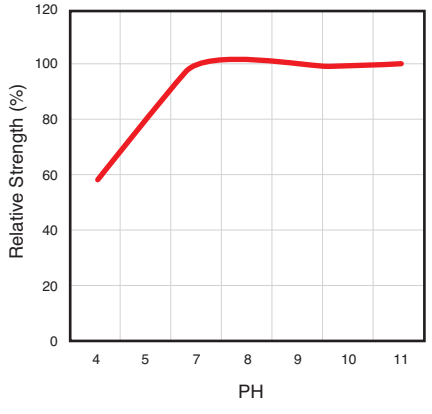
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	60
Cross-Dyeing 多種纖維沾染	Acetate	3-4
	Cotton	1
	Nylon	1
	Polyester	3-4
	Acrylic	3
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4B
Shade change by heat 熱變色性	0	3R
	2hr	4-5R

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Black HRN

Everlight Chemical

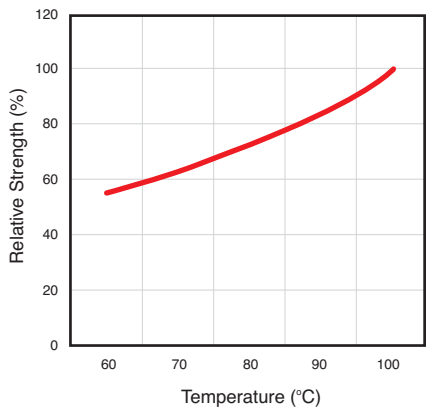
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		2.0%	4-5	5	3-4	5	5	5	5
	ISO C10 B		2.0%	4	5	2-3	5	5	5	5
Water 耐水	ISO 105-E01		2.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	2.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	2.0%	4-5						
		Wet 溼		2						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			2.0%	1-2R						
Colorfastness to Nitrogen Oxides 氣體變褪色			2.0%	4						

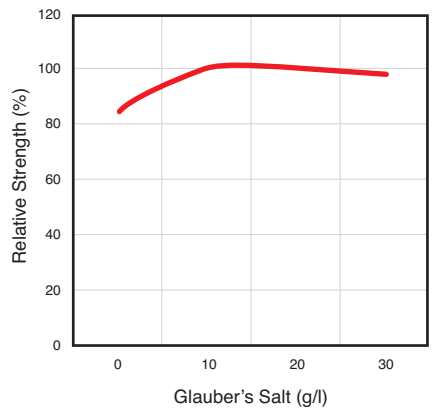
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	3-4
	0.1 %	1		Alkali		3-4
	0.5 %	1-2	Wet-Light 耐濕日光		0.2 %	1-2
	1.0 %	2-3			2.0 %	3-4
	2.0 %	3-4				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

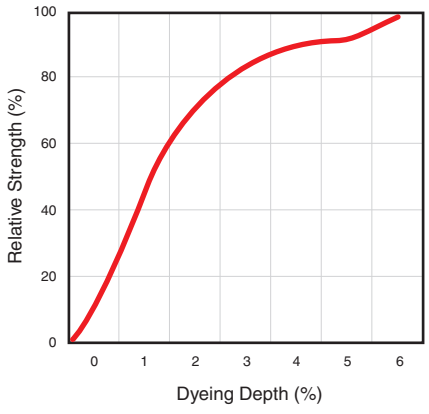


Everdirect Fast Black B160

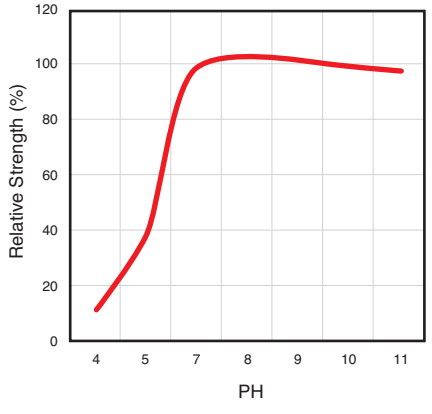
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		1-2
Solubility g/l 溶解度	80°C Water	20
Cross-Dyeing 多種纖維沾染	Acetate	3
	Cotton	1
	Nylon	1-2
	Polyester	3-4
	Acrylic	2-3
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4B
Shade change by heat 熱變色性	0	3R
	2hr	4R

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Fast Black B160

Everlight Chemical

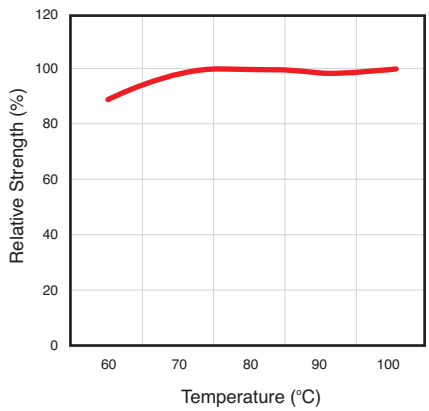
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		2.0%	4-5	5	3-4	5	5	5	5
	ISO C10 B		2.0%	4-5	5	3	5	5	5	5
Water 耐水	ISO 105-E01		2.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	2.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	2.0%	4-5						
		Wet 溼		2						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			2.0%	1-2YR						
Colorfastness to Nitrogen Oxides 氣體變褪色			2.0%	4-5						

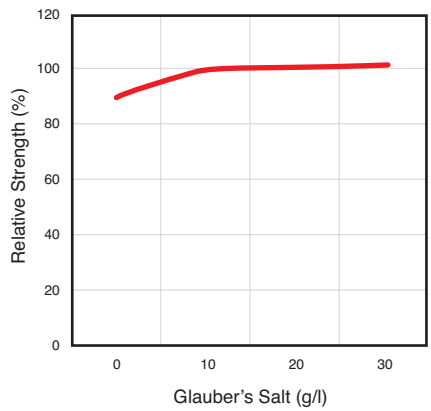
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	4
	0.1 %	1		Alkali		4
	0.5 %	2	Wet-Light 耐濕日光		0.2 %	1-2
	1.0 %	3			2.0 %	3-4
	2.0 %	4				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

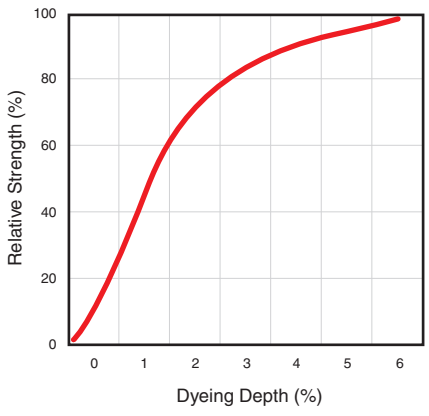


Everdirect Fast Black VSF 600

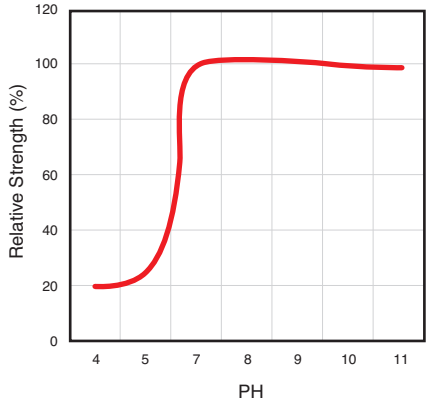
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		1-2
Solubility g/l 溶解度	80°C Water	30
Cross-Dyeing 多種纖維沾染	Acetate	3
	Cotton	1
	Nylon	1-2
	Polyester	3-4
	Acrylic	2-3
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5
	Cu^{2+} 銅	4-5
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	2-3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4B
Shade change by heat 熱變色性	0	3R
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Fast Black VSF 600

Everlight Chemical

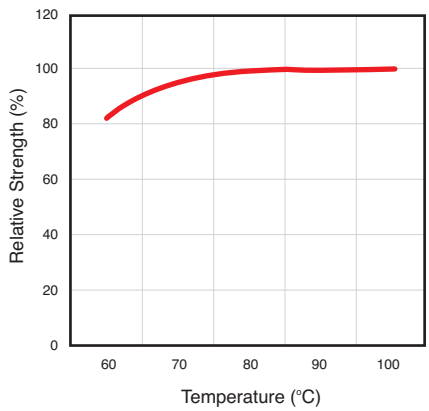
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		2.0%	4-5	5	3-4	5	5	5	5
	ISO C10 B		2.0%	4R	5	3	4	5	5	5
Water 耐水	ISO 105-E01		2.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	2.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	2.0%	4-5						
		Wet 溼		2						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			2.0%	2YR						
Colorfastness to Nitrogen Oxides 氣體變褪色			2.0%	4-5						

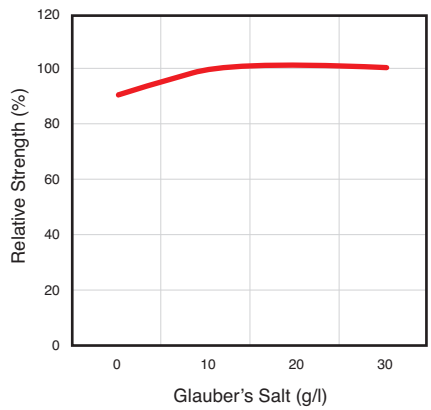
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	4
	0.1 %	1		Alkali		4
	0.5 %	2	Wet-Light 耐濕日光		0.2 %	1-2
	1.0 %	3			2.0 %	4
	2.0 %	4				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

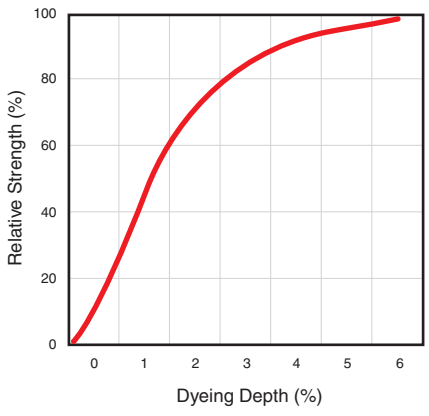


Everdirect Black ANBN H/C

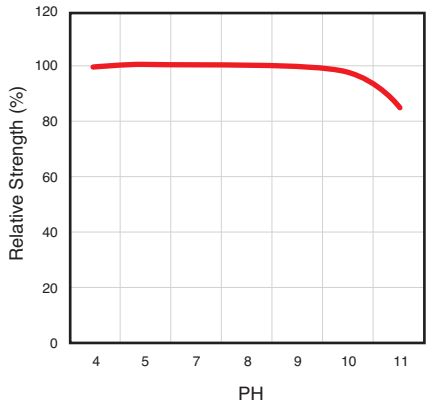
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		C
Exhaustion classification by Temp. 染色溫度分類		M
Migration Property 移染性		2
Solubility g/l 溶解度	80°C Water	40
Cross-Dyeing 多種纖維沾染	Acetate	1-2
	Cotton	1
	Nylon	1
	Polyester	1-2
	Acrylic	1-2
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5Y
	Cu^{2+} 銅	4Y
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4Y
Shade change by heat 熱變色性	0	3G
	2hr	4

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Black ANBN H/C

Everlight Chemical

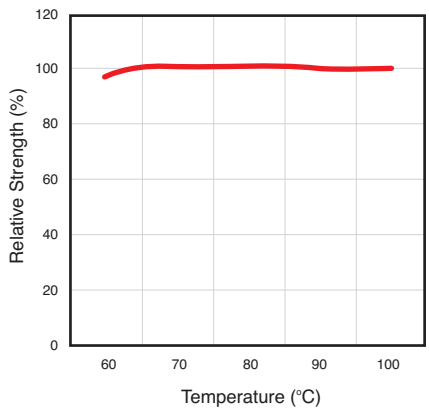
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		2.0%	4-5	4	1-2	2-3	4-5	4-5	5
	ISO C10 B		2.0%	4	3-4	1-2	3	5	4	4-5
Water 耐水	ISO 105-E01		2.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	2.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	2.0%	4-5						
		Wet 溼		2						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			2.0%	2YR						
Colorfastness to Nitrogen Oxides 氣體變褪色			2.0%	4-5						

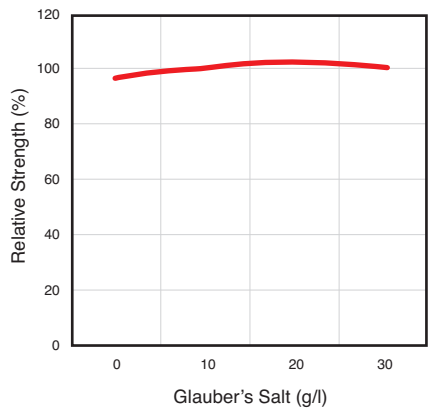
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	3
	0.1 %	1		Alkali		3
	0.5 %	1	Wet-Light 耐濕日光		0.2 %	1-2
	1.0 %	2			2.0 %	3Y
	2.0 %	3				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)

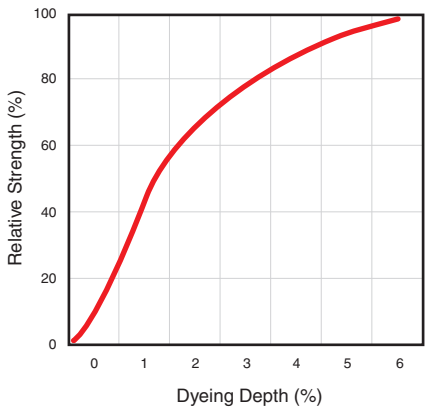


Everdirect Fast Black B300

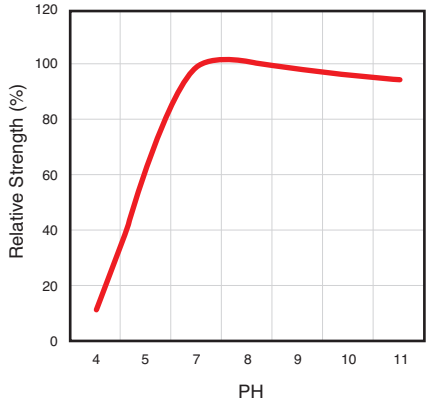
Dyeing Properties 特性

Characteristics		Evaluation
S.D.C. classification 分類		B
Exhaustion classification by Temp. 染色溫度分類		H
Migration Property 移染性		1-2
Solubility g/l 溶解度	80°C Water	40
Cross-Dyeing 多種纖維沾染	Acetate	1
	Cotton	1
	Nylon	1
	Polyester	2
	Acrylic	1-2
	Wool	1
Metallic ions 金屬離子	Fe^{2+} 鐵	4-5Y
	Cu^{2+} 銅	4BG
Color constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Discharge -ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3
Shade change by fixing agent 固色後變色性	Polyamine 聚胺類	4B
Shade change by heat 熱變色性	0	3R
	2hr	4-5

Build-up (染深色性)



Dependency on pH (pH依存性)



Everdirect Fast Black B300

Everlight Chemical

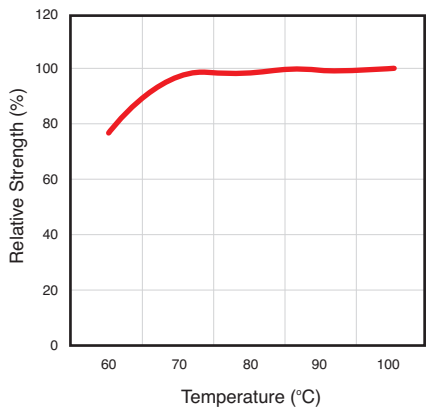
Fastness after fixing treatment 固色後牢度表

Evaluation / Fastness			Depth of shade 染色濃度	Color change 變褪色	Staining 污染					
					Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛
Washing 耐水洗	AATCC 61-2A		2.0%	4-5	5	3-4	5	5	5	5
	ISO C10 B		2.0%	4-5	4-5	3	4	5	5	5
Water 耐水	ISO 105-E01		2.0%	4-5	5	5	5	5	5	5
Perspiration 耐汗	ISO 105-E04	Acid 酸	2.0%	4-5	5	5	5	5	5	5
		Alkali 鹼		4-5	5	5	5	5	5	5
Rubbing 耐摩擦	ISO 105-X12	Dry 乾	2.0%	4-5						
		Wet 溼		2						
Chlorinated Water 耐氯水 (20ppm) ISO 105-E03			2.0%	2-3YR						
Colorfastness to Nitrogen Oxides 氣體變褪色			2.0%	4-5						

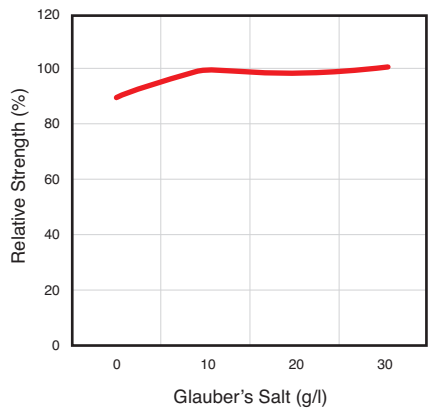
Light fastness before fixing treatment 固色前日光堅牢度

Evaluation	Dyeing Depth 染色濃度	Grade ISO Blue Scale	Evaluation		Dyeing Depth 染色濃度	Grade ISO Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.03 %	1	Perspiration Light 耐汗日光	Acid	2.0 %	4
	0.1 %	1		Alkali		4
	0.5 %	2	Wet-Light 耐濕日光		0.2 %	1-2
	1.0 %	3			2.0 %	4
	2.0 %	4				

Dependency on Temperature (溫度依存性)



Dependency on Glauber's (芒硝依存性)



Note

Note

本資料所提供的資訊與建議是以我們最大的用心完成，但不能含括所有可能的狀況。使用相關產品或搭配其他產品時，請先自行測試。本資料並不具保證及法律效用，僅供參考使用。使用者必須自行承擔所有的風險與法律責任。

臺灣永光化學工業股份有限公司－技術行銷處

Warnings： The information and recommendations presented here were completed with the utmost care, but can't be extended to cover every possible case. They are intended to serve as non-binding guidelines and must be adapted to the prevailing condition. The product whether used singly or in combination with other substances, should be carried out in lab test prior to bulk production. The user needs to assume all risk and liability no matter whatsover results are.

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